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Factor-augmented forecasting in big data☆

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ABSTRACT

This paper evaluates the predictive performance of various factor estimation methods in big data. Extensive forecasting experiments are examined using seven factor estimation methods with 13 decision rules determining the number of factors. The out-of-sample forecasting results show that the first Partial Least Squares factor (1-PLS) tends to be the best-performing method among all the possible alternatives. This finding is prevalent in many target variables under different forecasting horizons and models. This significant improvement can be explained by the PLS factor estimation strategy that considers the covariance with the target variable. Second, using a consistently estimated number of factors may not necessarily improve forecasting performance. The greatest predictive gain often derives from decision rules that do not consistently estimate the true number of factors.

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1. Introduction

A trend in economics is toward harnessing hundreds of economic time series. By introducing a small number of factors that govern comovement in data, parsimonious models can be achieved while preserving the important information. In this way, factor models have also enabled significant improvements in forecasting. Many studies have found encouraging empirical evidence for factor-augmented forecasts, also known as diffusion index forecasts. (Angelini, Camba-Mendez, Giannone, Reichlin, and Rünstler (2011), Artis, Banerjee, and Marcellino (2005), Barhoumi et al. (2008), Cristadoro, Forni, Reichlin, and Veronese (2005), Forni, Hallin, Lippi, and Reichlin (2003), Giannone, Reichlin, and Small (2008), Stock and Watson (1999, 2002a, 2002b), and Schumacher and Breitung (2008), among many).

There have been many attempts to estimate the latent factors, most commonly via Principal Components Analysis (PCA), which estimates the factors that best explain the variance of predictors. However, many alternative factor estimation methods aim to improve upon PCA. Although certain sets of factor estimation methods and decision rules to determine the number of factors are routinely used, large choice sets of possible factor-augmented forecasts are available.

Despite this variety, there is no consensus on their relative predictive ability. For instance, Schumacher (2007) finds that dynamic principal components outperform static principal components but are sensitive to parameter choice. Boivin and Ng (2005) find that the static principal components with unrestricted forecasts work systematically better in empirical analysis. D'Agostino and Giannone (2012) show that the two methods provide similar performance. Beyond principal components, comparing factor estimation methods is more challenging, as each method has different advantages and disadvantages. The question then arises as to how predictive these factor models are in typical big data. Further, what are their common empirical properties and what unexpected features should be considered in forecasting? More importantly, which factor

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estimation method and decision rule tend to provide the most accurate predictions?

Accordingly, this paper contributes to the literature by extensively comparing the performance of widely used factor estimation methods. First, our paper aims to combine various factor estimation methods, compare their predictive performance, propose the best-performing method, and serve as guideposts for forecasting macroeconomic and financial variables. Our findings show that the first Partial Least Squares factor (1-PLS) tends to be the best forecasting method. To our knowledge, this is a new finding in the forecasting literature. Second, this paper documents novel empirical properties and their implications for factor-augmented forecasts. For instance, we provide an empirical finding that consistently estimated numbers of factors may not always produce the best forecasts, which is particularly important given their popular use.

Specifically, seven factor estimation methods are tested: Principal Component Analysis, Weighted Principal Components (Boivin & Ng, 2006), one-sided estimation (Forni, Hallin, Lippi, & Reichlin, 2005), forecasting using targeted predictors (Bai & Ng, 2008), Partial Least Squares (Kelly and Pruitt (2015) and Ahn and Bae (2022)), two-step estimator (Doz, Giannone, & Reichlin, 2011), and quasi-maximum likelihood estimator (Doz, Giannone, & Reichlin, 2012). We construct a big dataset of 178 monthly U.S. macroeconomic and finance variables. Our dataset is in line with the conventional "big data" or so-called Stock–Watson data used in a vast macroeconomic literature, such as Bernanke and Boivin (2003), Bernanke, Boivin, and Eliasz (2005), McCracken and Ng (2016), Medeiros, Vasconcelos, Veiga, and Zilberman (2021), Stock and Watson (1996, 1998, 2002b) and Ludvigson, Ma, and Ng (2021) among many. In this extensive out-of-sample forecasting experiment, 148 target variables are forecasted using three forecasting equations across four forecasting horizons, using the seven factor estimation methods with 13 decision rules.

This paper reports two main findings. First, we find that Partial Least Squares (PLS) often outperforms other factor estimation methods: 1-PLS tends to show the largest predictive improvement and often becomes the best-performing method among all alternatives. The strong performance of 1-PLS is found in many target variables under various forecasting models and horizons. More specifically, 1-PLS is often the dominant decision rule for PLS to forecast a wide range of target variables, contrary to other factor estimation methods without strongly dominant decision rules. The strong predictive power of PLS derives from its factor estimation strategy. PLS estimates factors that have the maximum covariance with a target variable, which explains the significant forecasting improvement.

PLS was developed by Wold (1966, 1973, 1982) and the large-N and T properties of PLS factors are studied by Kelly and Pruitt (2015) and Groen and Kapetanios (2016). Recently, Ahn and Bae (2022) established the asymptotically optimal number of PLS factors under a general model and showed their general asymptotic and finite-sample properties. Lately, a growing body of literature analyses economic and finance variables using PLS; see Kelly and Pruitt (2013), Lin (2018) and Huang, Jiang, Tu, and Zhou (2015) for forecasting stock market returns, Zhang, He, Wang, and Liang (2022) for forecasting crude oil market volatility, Giglio, Kelly, and Pruitt (2016) and Bu, Rogers, and Wu (2021) for constructing systemic risk index and measure of monetary policy shocks, Light, Maslov, and Rytchkov (2017) for estimating expected returns on individual stocks, Hepenstrick and Marcellino (2019) and Marcellino and Sivec (2021) for nowcasting GDP growth using mixed-frequency data, Fuentes, Poncela, and Rodríguez (2015) for comparison of Sparse Partial Least Squares, PLS, and other alternatives including PCA. Despite this increasing popularity, it was not fully investigated how predictive PLS factors are in the typical big data, compared to a vast set of other widely-used factor estimation methods. This paper contributes to the existing literature by showing the strong performance of PLS in extensive out-of-sample forecasting experiments in the popular big data with macroeconomic and financial variables.

Second, a consistently estimated number of factors does not always lead to the best predictive performance in empirical practice. We examine the forecasting performance of 13 decision rules to determine the number of estimated factors. Many are consistent for the true number of latent factors that govern predictors, such as methods proposed by Alessi, Barigozzi, and Capasso (2010), Bai and Ng (2002, 2007), Hallin and Liška (2007), Onatski (2010) and Ahn and Horenstein (2013). However, decision rules that do not necessarily estimate the true number of factors consistently often provide better predictive accuracy. One possible explanation for this finding is the existence of irrelevant factors that govern predictors but do not strongly explain the target variable. The presence of irrelevant factors has been considered in the literature (Ahn and Bae (2022), Kelly and Pruitt (2015) and Huang, Jiang, Li, Tong, and Zhou (2022)). While the existing literature focuses on theoretical aspects of irrelevant factors, the novelty of our finding comes from providing empirically supporting evidence. In this sense, the first finding of encouraging forecasting performance of 1-PLS can also be explained by the fact that PLS can filter away the irrelevant factors that do not explain the target variable.

Also, another notable finding is that the determined numbers of factors vary significantly across decision rules, even among consistently estimated numbers of factors. Due to this reason, forecasting accuracy changes substantially among decision rules, even for the same factor estimation method. Therefore, choosing the factor estimation method and decision rule is crucial. However, despite this variety, 1-PLS tends to be the best-performing method among all the alternatives.

Some empirical evidence on the strong forecasting performance of PLS is proposed in the literature; see Ahn and Bae (2022) and Fuentes et al. (2015) among many others. While the former investigates the predictive performance of PLS on a set of macroeconomic variables to reassure their theoretical results, the latter compares the forecasting performance of PLS and Sparse Partial Least Squares. This paper aims to document the empirical properties of general factor-augmented forecasts, as well as

the forecasting behaviour of conventional decision rules. In addition, the relative forecasting performances across different factor estimation methods and decision rules are extensively compared.

This paper is organized as follows. Section 2 discusses the econometric framework and introduces factor estimation methods. Section 3 explains the data and forecasting experiments. Section 4 reports and interprets the empirical results, and Section 5 concludes. Throughout this paper, $\mathbf{1}_a$ denotes $a \times 1$ vector of ones. $P(B) = B(B'B)^{-1}B'$ and $Q(B) = I_a - P(B)$ for $a \times b$ full-column rank matrix B.

2. Forecasting model and factor estimation methods

2.1. Approximate dynamic factor model

We briefly introduce a simple approximate dynamic factor model that motivates our main forecasting framework. Let y_{t+1} be the one-period ahead forecast target being forecasted. The $N \times 1$ vector of predictors at time t is denoted as $x_t = (x_{1t}, \ldots, x_{Nt})'$. Both forecast target y_{t+1} and predictors x_t have mean zero for all $t = 1, \ldots, T$. The following dynamic factor model is assumed.

$$\begin{aligned} y_{t+1} &= \beta(L)f_t + \gamma(L)y_t + u_{t+1} \\ x_{it} &= \lambda_i(L)f_t + e_{it} \end{aligned} \tag{1}$$

where the dynamic factor f_t is a $q \times 1$ vector and $\beta(L)$, $\gamma(L)$ and $\lambda_i(L)$ are lag polynomials. More specifically, assuming $\beta(L)$ and $\lambda_i(L)$ having finite orders of at most s, the $1 \times q$ vectors $\beta(L)$ and $\lambda_i(L)$ are defined as $\beta(L) = \beta_0 + \beta_1 L + \cdots + \beta_s L^s$ and $\lambda_i(L) = \lambda_{i0} + \lambda_{i1}L + \cdots + \lambda_{is}L^s$ with $1 \times q$ vectors β_j and λ_j for $j = 0, 1, \ldots, s$. Then we can represent the above dynamic factor model (1) into a static form:

$$\begin{aligned} y_{t+1} &= \beta' F_t + \gamma(L)y_t + u_{t+1} \\ x_{it} &= \Lambda_i F_t + e_t, \end{aligned} \tag{2}$$

where $F_t = (f_t', \ldots, f_{t-s}')'$ and $\beta = (\beta_0, \ldots, \beta_s)'$ are $r \times 1$ vectors with $r = (s+1)q$ and $\lambda_i = (\lambda_{i0}, \ldots, \lambda_{is})$ is a $1 \times r$ vector. Alternatively, the model (3) can also be written as

$$\begin{aligned} y_{t+1} &= \beta' F_t + \gamma(L)y_t + u_{t+1} \\ x_t &= \Lambda F_t + e_t, \end{aligned} \tag{3}$$

where $e_t = (e_{1t}, \ldots, e_{Nt})'$ is a $N \times 1$ vector and $N \times r$ matrix Λ is defined as $\Lambda = [\Lambda_1', \ldots, \Lambda_N']'$. Letting $X = [x_1, \ldots, x_T]'$ and $y = [y_2, \ldots, y_{T+1}]'$ be the $T \times N$ matrix of predictors and $T \times 1$ vector of a target variable, our model is thus

$$\begin{aligned} y &= F\beta + \gamma(L)y + u \\ X &= F\Lambda' + E, \end{aligned} \tag{4}$$

where $F = (F_1, \ldots, F_T)'$, $u = (u_2, \ldots, u_{T+1})'$, and $E = (e_1, \ldots, e_T)'$.

Empirically, the static framework in (3) estimates factors by a time domain analysis, and the dynamic factors in model (1) are estimated by the frequency domain method. This paper also compares factor estimation methods based on both models (1) and (3).

The main empirical experiments focus on h-step forecasts in which the h-step-ahead projection is used directly to make the forecast. The multistep-ahead version of (3) gives the main econometric forecasting framework,

$$y_{t+h}^h = \delta_h + \beta_h(L)F_t + \gamma_h(L)y_t + u_{t+h}^h \tag{5}$$

where y_{t+h}^h is the h-step-ahead target variable. Note that the forecasting model (5) involves additional lags of F_t, denoted by $\beta_h(L)$. As the static representation of (2) shows, the static factors F_t already incorporate the past dynamic factors $f_t, \ldots, f_{t-s}$. However, many studies often incorporate the lagged static factors, denoted by $\beta_h(L)$ in (5) to detect possible effects of past static factors, such as Stock and Watson (2002b) and McCracken and Ng (2016), among many. Therefore, this paper also considers such a framework as one of the three main forecasting models in Section 3.2.

2.2. Factor estimation methods

We introduce a d dimensional factor G_t such that $\beta' F_t = \delta' G_t$ holds with a $d \times 1$ vector δ. Then, y_{t+1} in the model (1) and (3) can be replaced as follows.

$$y_{t+1} = \delta' G_t + \gamma(L)y_t + u_{t+1} \tag{6}$$

For forecasting, estimation of F_t is not essential, but rather $\beta' F_t$ which explains y_{t+1} is needed. Many factor estimation methods estimate $G_t = AF_t$ asymptotically, with an invertible $r \times r$ matrix A, which also makes $\beta' A^{-1} A F_t = \beta' F_t$. Then, $G_t = AF_t$, $\delta' = \beta' A^{-1}$ with $d = r$. However, some factor estimation methods, such as Partial Least Squares, can recover $\beta' F_t$ with G_t where $d \leq r$ (Ahn and Bae (2022), Kelly and Pruitt (2015), and Helland (1988, 1990)). To summarize, G_t are the factors being estimated; they do not need to be the same or have the same dimension as F_t, even asymptotically.

1. Principal Component Analysis (PCA)

Principal Components Analysis (hereafter "PCA") is popular, as discussed by Ahn and Horenstein (2013), Bai (2003), Bai and Ng (2002, 2006), Bernanke et al. (2005), Stock and Watson (2002a, 2002b, 2006) and Jurado, Ludvigson, and Ng (2015) among others. PCA estimates factor loadings Λ and factors $F_1, F_2, \ldots, F_T$ by solving the following least-squares problem,

$$\min_{\Lambda, \{F_t\}_{t=1}^T} \frac{1}{NT} \sum_{t=1}^{T} (x_t - \Lambda F_t)'(x_t - \Lambda F_t) \tag{7}$$

subject to a normalization of $N^{-1}\Lambda'\Lambda = I_r$. The solution, $\hat{\Lambda}^{PCA}$ reduces to the scaled r eigenvectors of the sample covariance matrix of predictors $\hat{\Sigma}_X = T^{-1}X'X$, corresponding to the largest r eigenvalues. The estimated PCA factors are $\hat{F}^{PCA} = N^{-1}X\hat{\Lambda}^{PCA}$.

2. Partial least squares (PLS)

PCA is an unsupervised method and estimates factors that explain predictors most, not the target variable of interest. In contrast, PLS described in Table 1 is a supervised method that estimates factors that have the maximum covariance with the target variable and hence can improve the forecasting performance of PCA.

Table 1
PLS iteration.

1.	j denotes the number of iterations.
2.	For $j = 1$, find α_1 such that $\alpha_1 = \underset{\alpha}{\text{argmax}} \frac{1}{T}\sum_{t=1}^{T}(\alpha' x_t y_{t+1})^2$ subject to $N^{-1}\alpha'\alpha = 1$. The first PLS factor is $\hat{F}_1^{PLS} = X\alpha_1 = XX'y$.
3.	For $j > 1$, update predictors to X^*, where $X_j^* = Q(\hat{F}_1^{PLS}, \ldots, \hat{F}_{j-1}^{PLS})X$ and $X_j^* = [x_{j,1}^*, \ldots, x_{j,T}^*]'$.
4.	Find α_j that is $\alpha_j = \underset{\alpha}{\text{argmax}} \frac{1}{T}\sum_{t=1}^{T}(\alpha' x_{j,t}^* y_{t+1})^2$, subject to $N^{-1}\alpha'\alpha = 1$. The jth PLS factor is $\hat{F}_j^{PLS} = X_j^*\alpha_j$. For the $j+1$ iteration, repeat Steps 3 and 4.

This table describes PLS iterations. Strictly, the first PLS factor in Step 2 should be $\hat{F}_1^{PLS} = X\alpha_1 = XX'y/\sqrt{y'XX'y}$ because $\alpha_1 = X'y/\sqrt{y'XX'y}$. Since the scalar in the denominator of α_1 is for normalization, $\sqrt{y'XX'y}$ does not affect projection. Therefore, the denominator is omitted for simplicity as it does not change forecasts.

PLS was first developed by Wold (1966, 1973, 1982). The large-N and large-T properties of PLS factors have been studied (Kelly and Pruitt (2015) and Groen and Kapetanios (2016)), and more general properties of PLS are documented by Ahn and Bae (2022). Recently, PLS has been popularly used in economics and finance; see Bu et al. (2021), Fuentes et al. (2015), Giglio et al. (2016), Hepenstrick and Marcellino (2019), Huang et al. (2015), Kelly and Pruitt (2013), Light et al. (2017), Lin (2018), Marcellino and Sivec (2021) and Zhang et al. (2022), among many.

Specifically, Ahn and Bae (2022) show the asymptotic properties of PLS in a general environment. They assume that $T^{-1}F'F = T^{-1}[F_{(1)}, \ldots, F_{(J)}]'[F_{(1)}, \ldots, F_{(J)}] \to_p diag(\sigma_1^2 I_{r(1)}, \ldots, \sigma_J^2 I_{r(J)})$ and $N^{-1}\Lambda'\Lambda \to_p I_r$, where for $j = 1, \ldots, J$, the $T \times r(j)$ matrix $F_{(j)}$ is defined as $F_{(j)} = [f_{(j),1}, \ldots, f_{(j),T}]'$ with $r(j)\times 1$ vector $f_{(j),t}$, $r = \sum_{j=1}^{J} r(j)$ and $\sigma_1^2 > \sigma_2^2 > \cdots > \sigma_J^2 > 0$. This assumption implies that r dimensional factors F consist of J groups of factors $\{F_{(j)}\}_{j=1}^{J}$, where the $r(j)$ dimensional subset of factors $F_{(j)}$ in the jth group shares the same asymptotic variance σ_j^2. Therefore, among r factors, only $J \le r$ factors have distinct asymptotic variance. They further generalize the assumption to $\beta_{(j)} = 0_{r(j)\times 1}$ for $j = (R+1), \ldots, J$, so that only the first R groups of factors affect the target variable.

Then $\beta' F_t$ in Eq. (3) is equivalent to $\mathbf{1}_R' G_t$, where the $R \times 1$ vector G_t is defined as $G_t = (f_{(1),t}'\beta_{(1)}, \ldots, f_{(R),t}'\beta_{(R)})'$, which implies $\delta = \mathbf{1}_R$ in Eq. (6). Ahn and Bae (2022) show that the first to Rth PLS factors can asymptotically span G_t, so that only the first R PLS factors are enough to recover $\beta' F_t$ where $R \le J \le r$. They further prove that using more than R PLS factors can reduce out-of-sample forecasting accuracy while inflating in-sample fit, even asymptotically. However, no method is known to estimate R consistently. The first PLS factor (1-PLS) tends to give the best result in their simulation results, even when $1 < R$. Therefore, 1-PLS, $\hat{F}_1^{PLS} = XX'y$, is explicitly considered here as a decision rule for PLS.

3. Targeted predictors

Bai and Ng (2008) propose using targeted predictors to improve forecasting performance, reasoning that PCA does not consider the predictive ability of each predictor x_{it} for the target variable in factor estimation. Accordingly, they propose first selecting a group of 'targeted' predictors with high predictive power for the target variable using LASSO (least absolute shrinkage and selection operator) or LARS (least angle regression). Second, PCA factors are estimated only from these targeted predictors. Similarly, the idea of combining factor estimation and shrinkage is also used in Kim and Swanson (2014) and Zhang, Wahab, and Wang (2023), among many others.

Generalized principal components (generalized PC)

Generalized principal components (hereafter "generalized PC") share the same intuition as generalized least squares (GLS). They can improve the efficiency of PCA, especially given heteroskedastic errors or idiosyncratic errors with cross-correlation. Let Σ_e be the true covariance matrix of idiosyncratic errors; then, generalized PC solves the weighted version of (7),

$$\min_{\Lambda, \{F_t\}_{t=1}^{T}} \frac{1}{NT}\sum_{t=1}^{T}(x_t - \Lambda F_t)'\Sigma_e^{-1}(x_t - \Lambda F_t) \tag{8}$$

subject to a normalization. The estimated factor loadings are the r scaled eigenvectors of $\Sigma_e^{-1/2}\hat{\Sigma}_X\Sigma_e^{-1/2'}$, corresponding to the r largest eigenvalues. The factors can be estimated by $\hat{F}^{GPC} = N^{-1}X\hat{\Lambda}^{GPC}$. Choi (2012) shows that generalized PC can provide a smaller forecasting error variance than PCA factors.

However, the above solution is infeasible because the true variance matrix of idiosyncratic errors Σ_e is unknown. A feasible generalized PC estimator is obtained by replacing Σ_e with the estimated $\hat{\Sigma}_e$. Two studies are considered that propose different $\hat{\Sigma}_e$, Boivin and Ng (2006) and Forni et al. (2005).

4. Weighted principal component (weighted PC)

Boivin and Ng (2006) propose generating a two-step diagonal weight matrix. First, PCA factors are estimated. Second, generalized PC factors are estimated that solve (8) with a diagonal matrix of $\hat{\Sigma}_e$, whose diagonal elements are the sample variance of estimated idiosyncratic error, $\hat{e}_t$ in the first step. A similar idea is also considered in Jones (2001).

5. One-sided estimation

Forni et al. (2005) propose one-sided estimation that uses the decomposition of the variance matrix of predictors, $\Sigma_X = \Sigma_{\Lambda F} + \Sigma_e$, where $\Sigma_{\Lambda F}$ is the variance of the common component from Eq. (3). This gives $\hat{\Sigma}_e = \hat{\Sigma}_X - \hat{\Sigma}_{\Lambda F}$, where $\hat{\Sigma}_{\Lambda F}$ is estimated by the dynamic principal component analysis of Forni, Hallin, Lippi, and Reichlin (2000), which is based on Eq. (1). More specifically, denote a consistent lag-window or periodogram-smoothing estimator of the $N \times N$ spectral density matrix of predictors $\Sigma(\theta)$ as $\hat{\Sigma}(\theta)$ for $\theta \in [-\pi, \pi]$. The first q eigenvalues and eigenvectors of $\hat{\Sigma}(\theta)$ provide estimates of the spectral density and variance matrix of the common component, $\Sigma_{\Lambda F}$. In this paper, we estimate $\hat{\Sigma}(\theta)$ as

$$\hat{\Sigma}(\theta) = \frac{1}{2\pi} \sum_{u=-M_T}^{M_T} w(M_T^{-1}u)\hat{\Gamma}_u e^{-iu\theta} \tag{9}$$

where w, $\hat{\Gamma}_u$ and $M_T > 0$ are a window function, the u-lag sample cross-covariance matrix and truncation parameter, respectively. We use the window size $M_T = T^{1/2}$ and the triangular window for w. The one-sided estimation is studied, developed and used widely, including forecasting economic variables; see Barigozzi and Hallin (2017, 2020), Forni, Giovannelli, Lippi, and Soccorsi (2018), Forni, Hallin, Lippi, and Zaffaroni (2015, 2017) and Trucíos, Mazzeu, Hotta, Pereira, and Hallin (2021).

Hybrid models

While maximum likelihood estimation (MLE) of dynamic factor models was applied to small models (Geweke (1977), Geweke and Singleton (1981), Sargent, et al. (1977), Watson and Engle (1983) and Stock and Watson (1989)), hybrid methods estimate factors by combining the efficiency improvement of the state space approach with PCA that can be used with many predictors. This approach can be implemented in real-time, and both cross-sectional and time-series averaging are considered in factor estimation, contrary to PCA, which solely considers cross-sectional averages. However, hybrid models rely on assumptions regarding factor structure and distribution of idiosyncratic errors because of the state space representation of factor models. Here, two hybrid estimation methods are considered, namely Doz et al. (2011) and Doz et al. (2012). A survey on the related literature in nowcasting can be found in Bańbura, Giannone, Modugno, and Reichlin (2013) and on factor estimation using Kalman filter and smoothing in Poncela, Ruiz, and Miranda (2021).

6. Two-step estimation

Doz et al. (2011) propose two-step estimation for the model in Giannone et al. (2008). First, PCA factors are obtained, and the model parameters are estimated using the PCA factors. Second, the factors are updated using the Kalman smoother. In this sense, the PCA factors and the corresponding parameters initiate the maximum likelihood estimation algorithm. The two-step estimation and framework of Giannone et al. (2008) are applied by Angelini et al. (2011), Bańbura and Rünstler (2011) and Hindrayanto, Koopman, and de Winter (2016).

7. Quasi-maximum likelihood estimation (QMLE)

Maximum likelihood estimation is not feasible for large-dimensional data since it involves estimating too many parameters. Doz et al. (2012) propose first estimating the factors by MLE, assuming an exact factor model with zero correlation between idiosyncratic errors, where MLE is feasible.

They show that the estimated factors are consistent for the true factor space, even when the true model is an approximate factor model. In this sense, their model is the quasi-maximum likelihood estimation (hereafter "QMLE") of White (1982). QMLE repeats the two-step estimation until convergence. Their approach has been widely applied and studied; see Banbura, Giannone, and Reichlin (2010), Bańbura and Modugno (2014) and Hindrayanto et al. (2016).

3. Data and forecasting procedure

3.1. Data description and transformation

This paper follows the forecasting schemes of Stock and Watson (2002b). The forecasting experiment mimics real-time forecasting of 148 monthly macroeconomic and financial target variables in the United States. Factors and parameters are estimated, and forecasts are made recursively at every time period.

The dataset covers monthly variables from 1959:01 to 2019:12, containing 178 variables from FRED-MD (McCracken & Ng, 2016), FRED and ISM (Institute for Supply Management). The data set includes eight major categories, namely output and income ("Output & Income") , labour market, housing, consumption, orders and inventories ("Consumption"), money and credit, interest and exchange rates ("Int & Exch Rates"), prices, and stock market: See Appendix B for additional detail.

Four forecasting horizons, $h = 1, 6, 12, 24$, are tested. The first out-of-sample forecast is made at 1970:01 for a h-period ahead target variable at 1970:01+h. The forecasts are obtained by regressing h-step-ahead target variable y_{t+h}^h on regressors at t, which may include factors, lagged factors, and lagged target variables. Some variables are transformed to be stationary. For instance, real variables are assumed to be I(1) in logarithms. Since industrial production (IP) is a real variable, the following transformation is applied.

$$y_{t+h}^h = (1200/h)\ln(\mathrm{IP}_{t+h}/\mathrm{IP}_t) \text{ and } y_t = 1200\ln(\mathrm{IP}_t/\mathrm{IP}_{t-1}) \tag{10}$$

In the main forecasting Eq. (5), y_{t+h}^h and y_t from (10) are used as the target variable and lagged dependent variable, respectively. In contrast, price-related variables are assumed to be I(2) in logarithms. For example, CPI and other nominal variables are transformed to construct the target and lagged variables:

$$y_{t+h}^h = (1200/h)\ln(\mathrm{CPI}_{t+h}/\mathrm{CPI}_t) - 1200\ln(\mathrm{CPI}_t/\mathrm{CPI}_{t-1}) \text{ and } y_t = 1200\Delta\ln(\mathrm{CPI}_t/\mathrm{CPI}_{t-1}) \tag{11}$$

Then, the transformed variables are standardized to have unit variance and mean zero. Finally, the data are screened for outliers: Any observations whose values exceed ten times the interquartile range from the median are treated as missing values. Factors are estimated only from the balanced panel with 108 predictors.

PLS and forecasting with targeted predictors require target variables while factors are estimated. However, some variables have missing values in their initial observations. Therefore, only variables with more than 80% of observations in the first factor estimation period are chosen as targets. Accordingly, there are 144 target variables for 1-, 6- and 12-month ahead forecasting, and 148 for 24-month ahead prediction.

3.2. Parameter estimation and model selection

At a given time $\bar{T}$, the following procedure yields $\hat{y}^h_{\bar{T}+h}$. First, a sequence of factors $\{\hat{F}_t\}_{t=1}^{\bar{T}}$ are estimated, using all the available sample data $\{x_t\}_{t=1}^{\bar{T}}$. Second, the estimated parameters, $\hat{\delta}_h$, $\hat{\beta}_h(L)$ and $\hat{\gamma}_h(L)$ are obtained by regressing y^h_{t+h} on regressors and estimated factors. The forecast for $y^h_{\bar{T}+h}$, $\hat{y}^h_{\bar{T}+h|\bar{T}}$ is constructed as $\hat{y}^h_{\bar{T}+h|\bar{T}} = \hat{\delta}_h + \hat{\beta}_h(L)\hat{F}_{\bar{T}} + \hat{\gamma}_h(L)y_{\bar{T}}$. Accordingly, differently estimated factors lead to different forecasts.

Regarding the parameter and factor estimation window, two forecasting schemes are examined: recursive and rolling estimation. Starting from 1970:01, both schemes forecast the h-period ahead target variable at 1970:01+h. First, the recursive scheme expands the estimation window every month. Second, the rolling scheme uses a fixed estimation window of 10 years, following Molodtsova and Papell (2009), Stock and Watson (2007), Swanson (1998) and Forni et al. (2018), among many. Forecasting results from both estimation schemes are reported.

Based on Eq. (5), three different forecasting equations are tested, where k, p and m denote the number of contemporaneous factors, lagged dependent variables and lagged estimated factors, respectively. First, the equation termed DI includes only k contemporaneous factors, where $1 \leq k \leq 12$, $m = 1$, and $p = 0$. The second forecasting equation is denoted DIAR, which combines DI forecast with the AR process. The DIAR model incorporates k dimensional contemporaneous factors and p lagged dependent variables. For the DIAR forecast, $m = 1$, $0 \leq p \leq 6$ is chosen by BIC, and $1 \leq k \leq 12$. The last forecast is DIAR-LAG, in which m lagged estimated factors are included in the DIAR model. In the DIAR-LAG model, $1 \leq m \leq 3$, $0 \leq p \leq 6$ are chosen by BIC, and $1 \leq k \leq 4$.

All three forecasting equations involve choosing the number of contemporaneous factors, k, in a certain range. Other parameters, such as m and p, are chosen by BIC if they are included. In this sense, decision rules are used to determine k. At each time period, the number of factors is determined by decision rules and updated recursively. This experiment shows the predictive power of decision rules.

3.3. Decision rules

In this paper, decision rules are the methods that determine the number of estimated factors for forecasting. This experiment incorporates decision rules popularly used in practice: Bai and Ng (2002) (hereafter "BN"), Onatski (2010) (hereafter "ON"), Alessi et al. (2010) (hereafter "ABC"), and Ahn and Horenstein (2013) (hereafter "AH"), Bayesian Information Criteria implemented by Stock and Watson (2002b) (hereafter "BIC"), as well as the supervised method proposed by Giovannelli and Proietti (2016) (hereafter "GP"). Since the BN estimator requires a penalty function, four penalty functions are tested. BN$-p1$, BN$-p2$, BN$-p3$, and BN$-BIC$ denote BN estimators, using IC_{p1}, IC_{p2}, IC_{p3}, and BIC_3 of Bai and Ng (2002), respectively. Also, ABC$-L$ and ABC$-S$ are ABC estimators with large and small windows. Finally, Holm's method is used for the GP estimator.

Since some factor estimation methods involve distinct decision rules, three decision rules are applied only to certain factor estimation methods. First, "1-PLS" is forecasting with only the first PLS factor ($k = 1$ for all time series), suggested by Ahn and Bae (2022). Second, the GP estimator can be applied to certain cases. First, the GP estimator is only used for PCA or PCA-related estimation that ensures orthogonality between estimated factors. That is, PCA, targeted predictors, and Weighted PC (Rule B) explained in Section 4. Second, the GP estimator is only used in DI and DIAR models, as the original work of Giovannelli and Proietti (2016) does not consider the DIAR-LAG model.

Third, one-sided estimation is applied in two versions: static with $k = q$ and dynamic with $q(s+1) \leq k$. Dynamic one-sided estimation needs information about the number of dynamic factors, q, and static factors, k. Bai and Ng (2007) (hereafter "BN2007") propose a methodology that simultaneously determines both q and k; hence, their method is only applied to dynamic one-sided forecasts. On the other hand, since the method of Hallin and Liška (2007) (hereafter "HL2007") estimates only q, both versions are tested. Denote the estimated number of dynamic factors by HL2007 as $\hat{q}_{HL2007}$. In the static model, we let $k = \hat{q}_{HL2007}$. Second, the dynamic version is assuming $s = 1$, so we determine k as $k = 2\hat{q}_{HL2007}$.

To summarize, BIC, BN, ON, ABC, and AH are applied to all the factor estimation methods except dynamic one-sided estimation. In contrast, 1-PLS is only applied to PLS. Also, GP is used only for PCA, Targeted Predictors and Weighted PC (Rule B) in DI and DIAR models. Finally, BN2007 and HL2007 are applied only to the one-sided estimation. Among all the decision rules, BN$-p1$, BN$-p2$, BN$-p3$, ON, ABC and AH estimators are consistent for the true number of factors r. Similarly, BN2007 and HL2007 are consistent for q (and r for BN2007, as it involves BN estimator for estimation of r). In contrast, 1-PLS, BN$-BIC$, BIC and GP may not be consistent. For instance, 1-PLS is inconsistent unless $r = 1$ or all the relevant factors that explain y have the same asymptotic variance.

4. Empirical results

This section reports and interprets empirical results. All reported results are obtained in the recursive scheme unless mentioned otherwise. Relative mean squared error (RMSE) is used to measure the forecasting accuracy of factor-augment forecasts. RMSE here is defined as the mean squared error (MSE) of a method relative to that of an autoregressive model where the lag p is determined by BIC (AR(BIC)).

$$RMSE(\text{method}) = \frac{MSE(\text{method})}{MSE(\text{AR(BIC)})} = \frac{\sum_{t=1970:01}^{2019:12-h}(y_{t+h}^h - \hat{y}_{t+h|t}^h)^2}{\sum_{t=1970:01}^{2019:12-h}(y_{t+h}^h - \tilde{y}_{t+h|t}^h)^2}$$

where $\tilde{y}_{t+h|t}^h$ is the forecast by AR(BIC) at time t. RMSE larger than one implies that an AR forecast of the given target variable is better than the factor-augmented forecast. Therefore, a lower RMSE indicates greater forecasting improvements over an AR forecast.

Next, Diebold and Mariano (1995) test (hereafter "DM") is implemented to compare the predictive accuracy difference between AR(BIC) and factor-augmented forecasts from a statistical point of view. The null hypothesis is that the given factor-augmented and AR(BIC) forecasts have equal forecast accuracy against the alternative of the factor-augmented forecast being more accurate than AR(BIC). The results are displayed in asterisks along with RMSEs. For ${}^*p < 0.1$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$, the null hypothesis of equal predictive ability between the two is rejected based on the DM statistics, implying the significance under $\alpha = 0.1$, 0.05 and 0.01, respectively.

Factor estimation methods are implemented as in the original work to the extent that the econometric model in this paper allows. For instance, Bai and Ng (2008) use soft-thresholding using the elastic net estimator of Zou and Hastie (2005), a convex combination of ridge and LASSO regression that satisfies the following equation.

$$\min_{\beta} \; RSS + \lambda_1 \sum_{j=1}^{N} |\beta_j| + \lambda_2 \sum_{j=1}^{N} \beta_j^2$$

where RSS is sum of squared residuals from a regression of y_{t+h}^h on all available regressors. If λ_1 is larger, the regression approaches LASSO, and when λ_2 is larger, it approaches ridge regression. Bai and Ng (2008) report $\lambda_2 = 1.5, 0.5, 0.25$. Following the original work, the three values of λ_2 are implemented here, using the elastic net estimator of Zou and Hastie (2005).

In contrast, Boivin and Ng (2006) try several weighting matrices used in (8). The two weighting matrices that outperform others are chosen here: Rule SWa and SWb. For Rule SWa, the weight matrix $\hat{\Sigma}_e$ is a diagonal matrix whose elements are the estimated variances of idiosyncratic errors, which is also considered in Jones (2001). For Rule SWb, the weight matrix is a diagonal matrix whose elements are the sample average of absolute values of estimated covariances between idiosyncratic errors. While the two rules focus on residuals, they also estimate PCA factors only from specific groups of predictors, not from the whole data. Four rules are tested, and the most predictive (Rule B) is used here. Rule B estimates PCA factors only from nominal variables, which corresponds to Category 5 (Money and Credit), 6 (Interest and Exchange Rates), and 7 (Prices) in this paper.[1] Therefore, the most predictive three rules in Boivin and Ng (2006) are implemented: Rule SWa and SWb for the weight matrix and Rule B that estimates PCA factors only from nominal variables.

4.1. Major findings

4.1.1. Strong forecasting performance of 1-PLS

This section suggests which factor estimation method and decision rule provide the best predictions. Overall, 1-PLS often yields the best forecasting performance among all possible alternatives.

First, as Ahn and Bae (2022) show theoretically and from simulation results, the number of PLS factors significantly affects forecasting performance, compared to other factor estimation methods such as PCA. Figs. 1 and 2 show how the number of PLS and PCA factors affects the mean RMSE over all the target variables, according to the forecasting horizon, forecasting model and forecasting scheme.

The figures calculate the mean RMSE of each k over all target variables. The figures reveal three implications for PLS. First, in contrast to PCA, PLS shows a more dramatic increase in Mean RMSE as the number of PLS factors increases in many situations. Also, 1-PLS tends to show the lowest mean RMSE than any other number of PLS factors. Therefore, the figures imply that the number of PLS factors should be chosen with care, supporting the importance of decision rules for PLS.

Second, 2-PLS generally performs better than 1-PLS in the DI model under a recursive scheme, especially for shorter forecasting horizons of $h = 1$ or 6. But even in this case, using three or more PLS factors shows worse forecasting performance. It is possible that 1-PLS may not be sufficient to capture enough dynamics of the target variable, especially when the lagged target variable is not incorporated (DI model) and on shorter forecasting horizons where dynamic matters ($h = 1$ or 6).

Third, when h is short, 1- or 2-PLS perform similarly across all models in rolling estimation. Moreover, 2-PLS slightly outperforms 1-PLS in the DIAR-LAG model at $h = 1$. But as the forecasting horizon h increases, overfitting of PLS becomes more severe on average, in contrast to recursive estimation, where overfitting of PLS becomes mitigated over h.

Considering that Figs. 1–2 highlight the importance of decision rules for PLS, Fig. 3 reports the performance of each decision rule for PLS and PCA from various perspectives. More specifically, Fig. 3 shows the percentage of

[1] Boivin and Ng (2005) estimate three PCA factors from nominal variables in Rule B estimation. However, the number of factors can also be determined by decision rules for Rule B estimation. Forecasting with targeted predictors and Rule B estimates PCA factors only from subsets of predictors. Therefore, when decision rules are applied for these methods, the number of factors is decided using only these subsets of predictors, not the whole data.

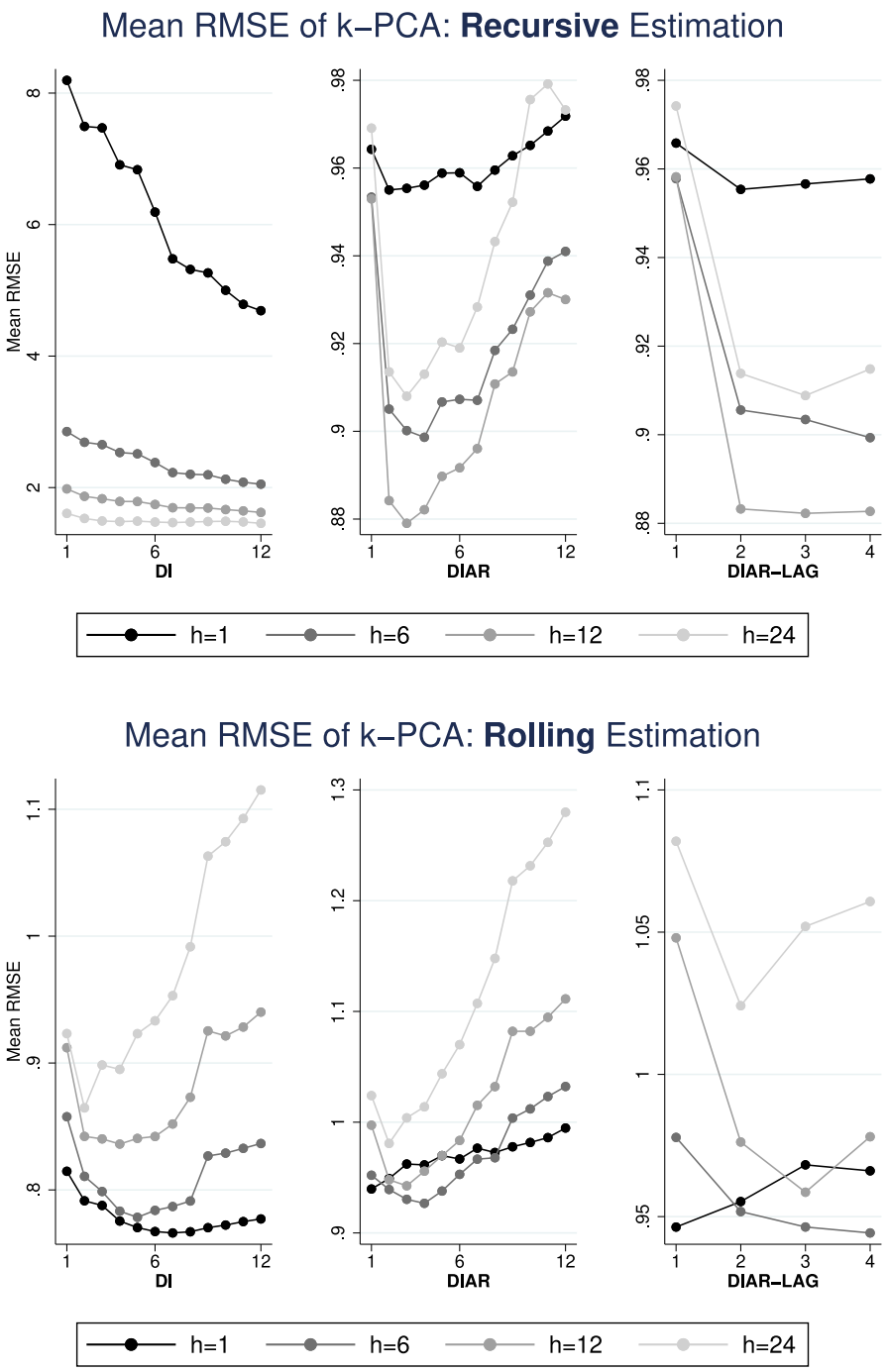


Fig. 1. Mean RMSE of k-PCA according to the forecasting scheme.

target variables that a decision rule outperforms all others for each forecasting model. The horizontal axis denotes the decision rules introduced in Section 3.3, and the vertical axis represents the percentage that the corresponding decision rule outperforms.

For PLS, the decision rule that gives the best results most frequently is shown by a dark red bar with the name of the decision rule bolded. For PCA, each forecasting model's three most dominant decision rules are shown in dark red bars with their names in bold. The figures imply that PCA does not have a strictly dominant decision rule for all forecasting models. For instance, $BN - BIC$ is the second best method in the DIAR model but not in the DI and DIAR-LAG model. $BN - p3$ exhibits good performance in the DI model but not in the DIAR and DIAR-LAG models. $BN - p1$ is the second-dominant decision rule in the DIAR-LAG model but not in the DI and DIAR models. However, 1-PLS is the dominant decision

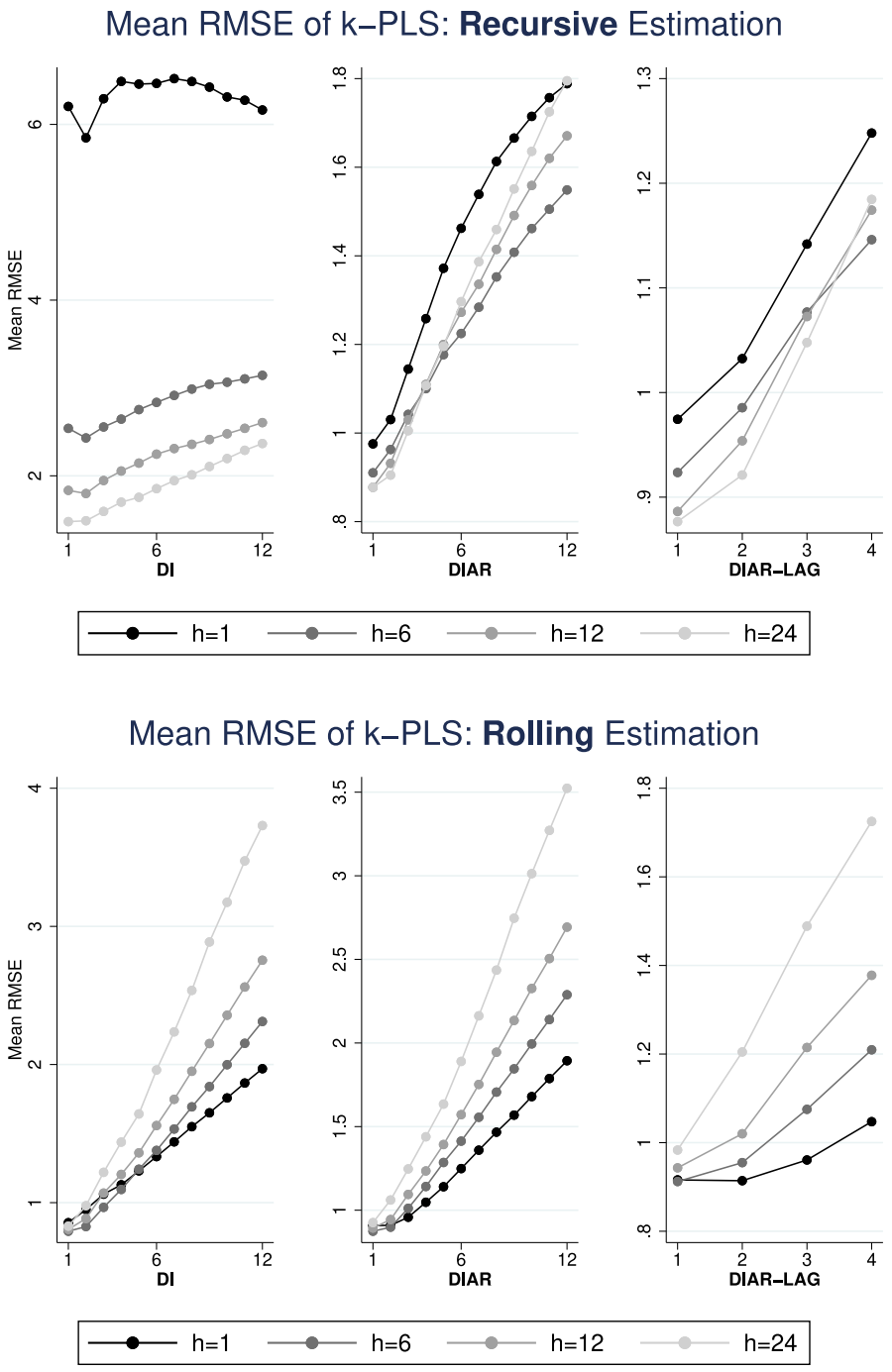
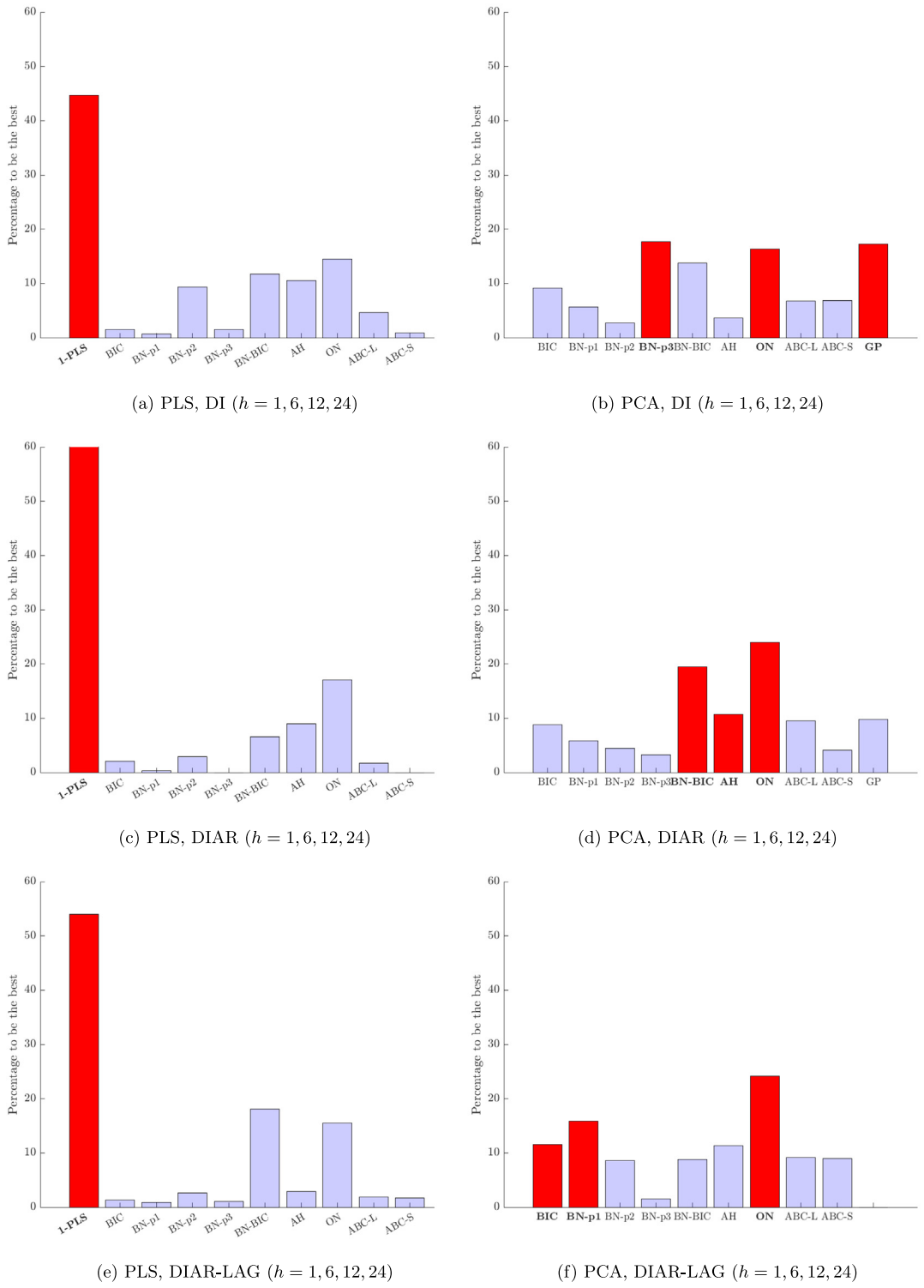


Fig. 2. Mean RMSE of *k*-PLS according to the forecasting scheme.

rule in all three forecasting models. For all models, 1-PLS gives the best results for around 50% of all target variables. Other decision rules can give better forecasting results than 1-PLS, but the difference is significant compared to the PCA case. Similar results are obtained under rolling estimation, demonstrated in Fig. 6 in Appendix A.

Next, Table 2 compares the relative forecasting accuracy of 1-PLS among all the possible alternatives for $h = 1, 6, 12, 24$. The left two columns of Table 2 show the top 5 methods, with decision rules in parentheses that perform best most frequently. On the other hand, the methods that perform closest to the best possible method on average are shown in the right two columns. The 'Ratio' of a given method represents $RMSE(Method)/RMSE(Best) - 1$, namely the percentage difference between the RMSEs of the best and the corresponding method. Therefore,



(a) PLS, DI ($h = 1, 6, 12, 24$)

(b) PCA, DI ($h = 1, 6, 12, 24$)

(c) PLS, DIAR ($h = 1, 6, 12, 24$)

(d) PCA, DIAR ($h = 1, 6, 12, 24$)

(e) PLS, DIAR-LAG ($h = 1, 6, 12, 24$)

(f) PCA, DIAR-LAG ($h = 1, 6, 12, 24$)

Fig. 3. Percentage of target variables that a decision rule outperforms over the rest of the decision rules using the forecasting model: PLS and PCA.

Mean(Ratio) denotes the average percentage difference between the best forecasts and the given method. Table 2 considers all target variables and forecasting models.

Table 2 documents that 1-PLS gives the best results most frequently in forecasting around 6%–24% of all target variables, as the left two columns show. The second-best

Table 2
Top 5 methods, PLS with other decision rules excluded.

	Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
	Methods	Percentage (%)	Methods	Mean(Ratio)
$h = 1$	**1-PLS**	**6.02**	OS-Static(BIC)	0.06
	Targeted Predictors, $\lambda = 1.5$(AH)	4.17	OS-Static(BN-p3)	0.062
	OS-Dynamic(HL2007)	3.47	Targeted Predictors, $\lambda = 1.5$(BIC)	0.065
	Weighted PC, Rule B(GP)	3.24	QMLE(BIC)	0.066
	PCA(GP)	2.55	Targeted Predictors, $\lambda = 0.5$(BIC)	0.069
			1-PLS	**0.232**
$h = 6$	**1-PLS**	**16.67**	Targeted Predictors, $\lambda = 1.5$(BIC)	0.1
	PCA(BN-BIC)	4.86	PCA(BIC)	0.103
	Targeted Predictors, $\lambda = 0.25$(BN-BIC)	3.24	OS-Static(BIC)	0.105
	Weighted PC, Rule B(GP)	3.24	Targeted Predictors, $\lambda = 0.5$(BIC)	0.105
	Weighted PC, Rule B(BIC)	3.01	Targeted Predictors, $\lambda = 0.25$(BIC)	0.107
			1-PLS	**0.142**
$h = 12$	**1-PLS**	**19.21**	**1-PLS**	**0.102**
	PCA(BN-BIC)	4.4	PCA(BN-BIC)	0.121
	OS-Dynamic(HL2007)	4.4	PCA(BIC)	0.121
	Weighted PC, SWa(AH)	3.24	PCA(ABC-L)	0.121
	Weighted PC, Rule B(BIC)	2.78	PCA(ABC-S)	0.124
$h = 24$	**1-PLS**	**24.1**	**1-PLS**	**0.076**
	Weighted PC, SWa(ON)	4.73	PCA(BN-BIC)	0.128
	Weighted PC, Rule B(AH)	2.7	PCA(ABC-L)	0.13
	Targeted Predictors, $\lambda = 0.5$(AH)	2.48	Weighted PC, SWb(ABC-L)	0.134
	Targeted Predictors, $\lambda = 1.5$(AH)	2.48	Two-step(BN-BIC)	0.136

This table presents the top 5 methods for each forecasting horizon giving the best results most frequently and giving results closest to the best results on average in the left and right columns respectively. 1-PLS is represented in bold and dashed line represents the case that 1-PLS is not the top 5 methods. PLS with other decision rules are excluded. For each forecasting horizon, all target variables in DI, DIAR and DIAR-LAG models are considered.

methods tend to vary depending on the forecasting horizon. Also, the results indicate that 1-PLS delivers the best forecasts more frequently and performs closer to the best methods as the horizons extend. For instance, 1-PLS is the best method in forecasting around 6% of target variables at $h = 1$. In this case, 1-PLS exhibits 23% larger RMSE than the best possible method, on average. While 1-PLS still provides the best results most frequently, Mean(Ratio) is relatively larger on a shorter horizon. This result implies that 1-PLS tends to perform best for many target variables in a shorter horizon but may have weaker forecasting results for others, increasing Mean(Ratio).

However, the Mean(Ratio) value of 1-PLS decreases with longer forecasting horizons. All methods except 1-PLS show higher 'Ratio' on average when forecasting long-run target variables. For instance, when forecasting 24-month ahead variables, 1-PLS is the best method for 24% target variables and shows only 7% larger RMSE than the best possible method. This can be explained by evidence that 1-PLS produces the best results more frequently given longer forecasting horizons. Since the best method has a zero Ratio value for the target variable, Mean(Ratio) decreases as 1-PLS becomes the best method more frequently. The same results can be found in Table 14 in Appendix A obtained under the rolling scheme: 1-PLS again remains one of the most dominating methods.

The predictive improvement of 1-PLS over longer h may be due to its supervised factor estimation strategy. Note that an unsupervised method, for instance, PCA, estimates factors regardless of the forecasting horizon h. As a result, even when h becomes larger, the factors used for forecasting remain the same for unsupervised methods. However, since PLS is estimated by the covariance between the target y_{t+h}^h and predictors x_t, as h extends, the relative predictive gain of PLS can become larger.

More detailed results of the relative forecasting accuracy of 1-PLS are presented in Tables 15–19 in Appendix A. Tables 3 and 4 decompose the forecasting results across the eight categories, focusing on 1-PLS. More specifically, the tables break down the results of 1-PLS by category that a target variable belongs to. While Table 3 includes PLS with other decision rules than 1-PLS, Table 4 excludes PLS with other decision rules. All forecasting horizons, target variables and forecasting models are considered. The percentage columns present the percentage of target variables in a given category in which 1-PLS achieves the best results among all the alternatives. The Mean(Ratio) reports the average predictive difference between the best possible method and 1-PLS in the category.

Table 3 documents that 1-PLS delivers the best forecasts more frequently for real variables under the recursive scheme. For example, 1-PLS provides the best prediction for around 26% of the target variables in the Output and Income category. On the other hand, 1-PLS tends to show better forecasting results for nominal variables under the rolling scheme. When a PLS with different decision rules than 1-PLS is excluded, the relative improvement of 1-PLS over the remaining competing methods becomes more evident in Table 4. Predictive improvement is greater for nominal variables. For instance, 1-PLS delivers the best results in forecasting 50% of price-related variables under the rolling scheme.

Table 3
1-PLS & best possible results by category, PLS with all decision rules.

	Recursive		Rolling	
	Percentage (%)	Mean(Ratio)	Percentage (%)	Mean(Ratio)
Overall	11.15	0.15	9.71	0.26
1. Output & Income	25.93	0.06	17.13	0.13
2. Labor Market	13.24	0.16	8.58	0.24
3. Housing	20.93	0.63	10.08	1.07
4. Consumption	7.92	0.15	3.75	0.31
5. Money and Credit	5	0.08	9.44	0.13
6. Int & Exch Rates	7.54	0.11	14.68	0.19
7. Prices	3.79	0.09	4.55	0.15
8. Stock Market	0	0.08	17.65	0.08

This table breaks down the results of 1-PLS by category that a target variable belongs to, according to forecasting scheme. The Percentage represents the percentage of target variables that 1-PLS achieves the best results among all the alternatives in a given category. Mean(Ratio) reports the average predictive difference between the best possible method and 1-PLS. PLS with other decision rules are all considered. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

Table 4
1-PLS & best possible results by category, PLS with other decision rules excluded.

	Recursive		Rolling	
	Percentage (%)	Mean(Ratio)	Percentage (%)	Mean(Ratio)
Overall	16.55	0.14	29.54	0.21
1. Output & Income	39.35	0.04	32.41	0.1
2. Labor Market	18.38	0.13	37.25	0.17
3. Housing	27.13	0.61	12.4	1.06
4. Consumption	13.33	0.14	11.67	0.3
5. Money and Credit	7.22	0.07	24.44	0.07
6. Int & Exch Rates	9.13	0.11	26.19	0.15
7. Prices	9.47	0.07	47.35	0.05
8. Stock Market	0	0.07	25.49	0.07

This table breaks down the results of 1-PLS by category that a target variable belongs to, according to forecasting scheme. The Percentage represents the percentage of target variables that 1-PLS achieves the best results among all the alternatives in a given category. Mean(Ratio) reports the average predictive difference between the best possible method and 1-PLS. PLS with other decision rules are excluded. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

Next, the predictive performance of 1-PLS is examined in Table 5 from a statistical point of view. The table lists how often 1-PLS forecasts show significant DM statistics under different significance levels α, for $\alpha = 0.1, 0.05$ and 0.01, by forecasting scheme and target variable's category. For all α, DM statistics of 1-PLS under the rolling scheme are more often significant for variables in the labour market (2. Labor Market) and housing (3. Housing). Also, rolling estimation demonstrates improvement in forecasting nominal variables: 5. Money and Credit, 7. Prices and 8. Stock Market. The improvement can also be attributed to a reduction in parameter instability. For instance, the parameter instability of economic variables is studied by many works, including but not limited to Giacomini and White (2006), Stock and Watson (1996, 2007, 2009) and Inoue, Jin, and Rossi (2017), among others. Since nominal variables are more prone to instabilities, the forecasting improvement of 1-PLS in rolling estimation can be explained.

Finally, Tables 6–7 compare 1-PLS and PCA. These results are particularly important given the widespread use of PCA. Only PCA and 1-PLS are considered here. Table 6 compares the predictive ability of 1-PLS and k-PCA for all target variables, according to the forecasting model, where k is fixed as $1 \leq k \leq k_{max}$. Each forecasting model considers all target variables and forecasting horizons, giving the 580 variable-horizon combinations. Frequency counts how often the given method achieves the minimum RMSE among the 580 combinations. The percentage is the frequency divided by 580 and multiplied by 100. Table 6 demonstrates that 1-PLS outperforms other PCA forecasts with given $k = 1, 2, \ldots, k_{max}$, around 230 times out of 580 combinations (around 40%). Similarly, Table 7 compares 1-PLS and PCA with decision rules, from PCA : *BIC* to PCA : *GP* by forecasting model. Again, the table shows that 1-PLS dominates PCA forecasts with ten decision rules, around 210 times out of 580 combinations on average.

The Fluctuation test proposed by Giacomini and Rossi (2010) is implemented to compare predictive accuracy between 1-PLS and PCA in the presence of possible instabilities. This test analyzes the change in relative performance between 1-PLS and PCA over time in the sample

Table 5
Significant DM statistics of 1-PLS by category.

	Method	Recursive		Rolling	
		Frequency	Percentage (%)	Frequency	Percentage (%)
$\alpha = 0.1$	Overall	597	34.31	823	47.3
	1. Output & Income	153	70.83	125	57.87
	2. Labor Market	218	53.43	306	75
	3. Housing	18	13.95	30	23.26
	4. Consumption	76	31.67	70	29.17
	5. Money and Credit	22	12.22	55	30.56
	6. Int & Exch Rates	94	37.3	70	27.78
	7. Prices	14	5.3	154	58.33
	8. Stock Market	2	3.92	13	25.49
$\alpha = 0.05$	Overall	429	24.66	682	39.2
	1. Output & Income	109	50.46	83	38.43
	2. Labor Market	173	42.4	291	71.32
	3. Housing	11	8.53	22	17.05
	4. Consumption	48	20	51	21.25
	5. Money and Credit	11	6.11	44	24.44
	6. Int & Exch Rates	70	27.78	52	20.63
	7. Prices	7	2.65	131	49.62
	8. Stock Market	0	0	8	15.69
$\alpha = 0.01$	Overall	193	11.09	486	27.93
	1. Output & Income	38	17.59	43	19.91
	2. Labor Market	116	28.43	242	59.31
	3. Housing	1	0.78	9	6.98
	4. Consumption	12	5	35	14.58
	5. Money and Credit	4	2.22	33	18.33
	6. Int & Exch Rates	20	7.94	31	12.3
	7. Prices	2	0.76	90	34.09
	8. Stock Market	0	0	3	5.88

This table breaks down the DM statistics results of 1-PLS by category that a target variable belongs to, according to significance level (α) and forecasting scheme. The Frequency represents how often 1-PLS shows significant DM statistics in forecasting a target variable in a given category. The Percentage divides the Frequency by the number of target variables belong to the category. Note that summing all values in the frequency columns over the eight categories is identical to the frequency in 'Overall'. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

under the rolling scheme. Fig. 4 reports the (Giacomini & Rossi, 2010) Fluctuation test on 1-PLS and PCA with the best possible decision rule for forecasting 1. Industrial Production and 2. CPI at $h = 1$ and $h = 12$ in rolling estimation.

The blue line of G-R test statistics graphically demonstrates the relative measure of local performance over the sample path. The null hypothesis is that the two models, 1-PLS and PCA, perform equally well at each point of time, against the alternative that 1-PLS outperforms PCA at some point in time. When the test statistic in the blue line is above the red critical value line, 1-PLS significantly perform better than PCA forecasts in Fig. 4.

Several empirical implications can be drawn from the figures. First, we are not able to reject the null hypothesis in forecasting the Industrial Production during the forecasting period for $h = 1$ and $h = 12$. In contrast, 1-PLS performs significantly better forecasting CPI than PCA(Best). Especially when forecasting 12 months ahead of CPI, 1-PLS shows considerable improvement upon PCA(BEST) for almost all periods. This finding also aligns with Fuentes et al. (2015), which indicates that PLS factors tend to forecast inflation based on CPI better than PCA or Targeted Predictor forecasts.

Why does 1-PLS perform better than PCA? Again, it can be explained by its supervised method that estimates the factors with the highest covariance with the target variable. To illustrate this difference in detail, Fig. 5 shows the contribution of each predictor to 1-PLS and PCA factors in heat maps. Specifically, the two graphs show the squared loadings for k-PCA and 1-PLS, where $k = 1, \ldots, 12$. Since PLS factors and loadings vary between forecasting horizons h and target variables, we select eight target variables in every category from 1. Real Personal Income to 8. S&P 500. The number next to the target variable's name denotes the category that it belongs to. While the top graph shows the squared loadings of k-PCA, the graph at the bottom of Fig. 5 illustrates the squared loadings of 1-PLS of the eight variables for $h = 12$.

Fig. 5 shows that 1-PLS and PCA loadings can be clearly distinguished. First, real variables in the first four categories mainly load on the first PCA factor. In contrast, the second, third, fourth and fifth PCA factors have the largest loadings on variables belong to 6. Interest and Exchange Rates, 7. Prices, 8. Stock Market and 1. Output & Income categories, respectively. However, the second graph of Fig. 5 shows that the 1-PLS factors of certain target variables often have the largest loadings on predictors from the same category. For instance, the target variables of Nonagriculture Employment, Housing starts, Monetary Base, 10-Year Treasury Rate, CPI and S&P 500 are explained more by predictors from the same category.

Table 6
1-PLS & k-PCA.

	DI		DIAR		DIAR-LAG	
	Frequency	Percentage (%)	Frequency	Percentage (%)	Frequency	Percentage (%)
1-PLS	**246**	**42.41**	**209**	**36.03**	**230**	**39.66**
1-PCA	59	10.17	88	15.17	99	17.07
2-PCA	19	3.28	33	5.69	74	12.76
3-PCA	30	5.17	46	7.93	72	12.41
4-PCA	36	6.21	62	10.69	105	18.1
5-PCA	20	3.45	28	4.83		
6-PCA	14	2.41	26	4.48		
7-PCA	30	5.17	35	6.03		
8-PCA	11	1.9	8	1.38		
9-PCA	13	2.24	21	3.62		
10-PCA	12	2.07	6	1.03		
11-PCA	23	3.97	3	0.52		
12-PCA	67	11.55	15	2.59		

1-PLS and k-PCA, $k = 1, 2, \ldots, 12$, for all target variables in 1-, 6-, 12- and 24-month ahead DI, DIAR and DIAR-LAG forecasting are considered. For DIAR-LAG, the maximum k is 4, so DIAR-LAG k-PCA forecasts with $4 < k$ are left empty. The frequency is how often the given method achieves the minimum RMSE in 580 variable-horizon combinations. The percentage is the frequency divided by 580 and multiplied by 100.

Table 7
1-PLS & PCA with decision rules.

	DI		DIAR		DIAR-LAG	
	Frequency	Percentage (%)	Frequency	Percentage (%)	Frequency	Percentage (%)
1-PLS	**216**	**37.24**	**202**	**34.83**	**198**	**34.14**
$PCA : BIC$	37	6.38	28	4.83	48	8.28
$PCA : BN - p1$	25	4.31	30	5.17	62	10.69
$PCA : BN - p2$	14	2.41	15	2.59	35	6.03
$PCA : BN - p3$	70	12.07	9	1.55	5	0.86
$PCA : BN - BIC$	34	5.86	77	13.28	38	6.55
$PCA : AH$	12	2.07	42	7.24	50	8.62
$PCA : ON$	42	7.24	81	13.97	75	12.93
$PCA : ABC - L$	27	4.66	41	7.07	38	6.55
$PCA : ABC - S$	27	4.66	20	3.45	31	5.34
$PCA : GP$	76	13.1	35	6.03		

1-PLS and PCA with decision rules, for all target variables in 1-, 6-, 12- and 24-month ahead DI, DIAR and DIAR-LAG forecasting are considered. The frequency is how often the given method achieves the minimum RMSE in 580 variable-horizon combinations. The percentage is the frequency divided by 580 and multiplied by 100. As GP is not used in DIAR-LAG model, the corresponding entries remain empty.

Second, it is interesting that the 1-PLS loadings reveal economic relationships between several variables. For instance, Real Consumption has the highest loadings from variables in 6. Interest and Exchange Rates: this observation is consistent with the well-known relationship between real interest rates and household consumption patterns. Also, the CPI has the highest loadings from the Real M2 Money Stock (M2REAL), as expected. Therefore, Fig. 5 emphasizes the differences in factor estimation strategies between PLS and PCA and hence explains the stronger forecasting performance of PLS.

To summarize, 1-PLS shows the strongest forecasting performance among all the possible alternatives. The significant predictive accuracy of PLS in these extensive experiments can be explained by its factor estimation strategy that estimates factors having the maximum covariance with a target variable, as explained in Table 1.[2]

[2] Then why the first PLS factor, in particular, shows the best forecasting performance than any other number of the PLS factors? Ahn and Bae (2022) gives some possible explanation of this finding as the ill-conditioned properties of Vandermonde matrices. They show that theoretically, the first R number of PLS factors asymptotically estimates G_t in Section 2.2, but under rotation of Vandermonde matrices. It is well known that the Vandermonde matrices are ill-conditioned (see Dax (2017)), implying that in the finite sample, the first PLS factor may explain most of the forecasting power that the rest of the PLS factors have.

4.1.2. Decision rules and forecasting performance

It may be unclear how many contemporaneous factors should be used in forecasting. Using the available data, researchers should often decide the number of factors according to the research design. Accordingly, this section evaluates the performance of decision rules introduced in Section 3.3.

The forecasting results with all the decision rules and factor estimation methods are summarized in Tables 8 to 9 for the eight variables in Stock and Watson (2002b) by forecasting horizons. The RMSE panel compares the RMSEs between the best method and 1-PLS. For $^*p < 0.1$,

Table 8
DIAR forecasts, real variables.

Industrial Production	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.86	0.727		
	1-PLS	0.876	0.745	0.684	0.679
	Name: Best Method				
		TP, $\lambda = 0.5$ (BN-BIC)	PCA (ON)	1-PLS	1-PLS
Personal Income	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.943			0.812
	1-PLS	0.952	0.771	0.7	0.818
	Name: Best Method				
		PCA (BN-BIC)	1-PLS	1-PLS	PLS (ON)
Mfg & Trade Sales	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.9	0.679	0.623	0.703
	1-PLS	0.932	0.683	0.646	0.77
	Name: Best Method				
		PCA (AH)	PLS (ON)	WPC, SWa (ON)	WPC, SWa (ON)
Nonag. Employment	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best			0.768	0.768
	1-PLS	0.848	0.792	0.79	0.779
	Name: Best Method				
		1-PLS	1-PLS	PCA (BN-BIC)	OS-S (BIC)

Tables 8–9 summarize the forecasting accuracy of all factor estimation methods with decision rules by forecasting horizons. The RMSE panel shows the relative mean squared errors (RMSE) of the best possible forecast and 1-PLS. The Name panel presents the name of the best method. The eight target variables, 1. Industrial Production to 8. Producer Price Index, are forecasted by h-month-ahead DIAR forecasting. For $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$, the null hypothesis of equal predictive ability between 1-PLS and best method is rejected based on Diebold–Mariano test.

$^{**}p < 0.05$, $^{***}p < 0.01$, the null hypothesis of equal predictive ability between 1-PLS and the best forecast is rejected based on Diebold–Mariano test. The Name panel shows the name of the best method.

The two tables show that 1-PLS provides the best forecasts most often, in 25% of the variable-horizon combinations. Even when 1-PLS does not produce the best results, the predictive difference between the best method and 1-PLS is usually not statistically significant. Second, the two tables imply that even when we restrict our attention to the eight variables, the same factor estimation method can provide fairly different forecasts depending on the decision rules.

Additionally, the determined number of factors varies significantly across decision rules. Table 10 reports summary statistics of the determined number of static and dynamic factors, $\hat{k}$ and $\hat{q}$, by decision rules. The table shows the mean, standard deviation, maximum, and minimum of recursively determined $\hat{k}$ and $\hat{q}$. $\hat{k}$ varies significantly across decision rules, from an average minimum of 1.34 to a maximum of 12. Even when the same decision rule is applied, different specifications of the rule, such as the choice of a penalty function ($\hat{k}_{BN}$) or tuning parameter ($\hat{k}_{ABC}$), results in quite different numbers of factors. The number of factors also widely differs among decision methods that consistently estimate the true number of factors.

Tables 11 and 12 describe the relative forecasting performances of decision rules for a given factor estimation method. In general, no specific decision rule tends to deliver the best results for many target variables for most factor estimation methods. Table 11 reports the percentage of target variables that a decision rule outperforms all the rest of the decision rules for PCA by category. For example, $\hat{k}_{ON}$ yields the best results for 38.89% of target variables in the output and income category when PCA factors are used for forecasting. $\hat{k}_{ON}$ also outperforms in forecasting consumption and stock market variables. However, $\hat{k}_{AH}$ appears to dominate in the housing category, $\hat{k}_{BN-BIC}$ in variables related to the labour market and $\hat{k}_{ABC-L}$ in interest or exchange rate variables. $\hat{k}_{GP}$ and $\hat{k}_{BIC}$ perform well in forecasting price-related variables and money and credit variables, respectively.

When taken together, the results may suggest that a target variable can be governed by a subset of factors that affect predictors. These relevant factors can vary across target variables and may not always best explain predictors, as Kelly and Pruitt (2015) and Ahn and Bae

Table 9
DIAR forecasts, price-related variables.

CPI	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.939	0.966		0.781
	1-PLS	0.998	0.972	0.883	0.796
	Name: Best Method				
		TP, $\lambda = 1.5$ (BN-BIC)	PLS (ON)	1-PLS	TP, $\lambda = 1.5$ (AH)
Consumption Deflator	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.959***	1.016	0.968	0.842
	1-PLS	1.035	1.019	0.974	0.88
	Name: Best Method				
		TP, $\lambda = 1.5$ (BN-BIC)	PLS (BN-BIC)	WPC, SWa (BN-BIC)	OS-S (BN-BIC)
CPI exc. Food	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.951		0.899	0.829
	1-PLS	1.019	0.98	0.91	0.839
	Name: Best Method				
		WPC, SWb (BN-p1)	1-PLS	TP, $\lambda = 0.25$ (AH)	PLS (AH)
Producer Price Index	RMSE: Best vs 1-PLS				
		$h = 1$	$h = 6$	$h = 12$	$h = 24$
	Best	0.963***	0.993	0.959	0.879
	1-PLS	1.013	1.028	0.995	0.928
	Name: Best Method				
		TP, $\lambda = 0.5$ (AH)	WPC, Rule B (BN-p2)	PCA (GP)	WPC, Rule B (BN-BIC)

Table 10
Statistics of number of factors determined by decision rules.

	Mean	Std	Max	Min		Mean	Std	Max	Min
$\hat{k}_{BN-p1}$	8.7	1.19	11	4	$\hat{k}_{ON}$	1.34	0.49	3	1
$\hat{k}_{BN-p2}$	6.91	2.24	10	2	$\hat{k}_{ABC-L}$	4.96	0.87	8	2
$\hat{k}_{BN-p3}$	12	0	12	12	$\hat{k}_{ABC-S}$	8.19	0.87	11	5
$\hat{k}_{BN-BIC}$	3.3	0.6	4	2	$\hat{q}_{BN2007}$	5.25	1.5	7	2
$\hat{k}_{AH}$	1.72	0.45	2	1	$\hat{q}_{HL2007}$	2.37	0.56	4	1

Statistics of all decision rules are obtained from DIAR model. $\hat{q}_{BN2007}$ and $\hat{r}_{BN2007}$ denote the estimated number of dynamic and static factors, q and r by Bai and Ng (2007). Note that $\hat{r}_{BN2007}$ is not displayed because $\hat{r}_{BN2007} = \hat{k}_{BN-p2}$.

Table 11
Percentage of target variables that a decision rule outperforms the rest decision rules: PCA.

	$\hat{k}_{BIC}$	$\hat{k}_{BN-p1}$	$\hat{k}_{BN-p2}$	$\hat{k}_{BN-p3}$	$\hat{k}_{BN-BIC}$	$\hat{k}_{AH}$	$\hat{k}_{ON}$	$\hat{k}_{ABC-L}$	$\hat{k}_{ABC-S}$	$\hat{k}_{GP}$
1. Output & Income	5.56	9.26	5.09	6.02	15.28	6.48	**38.89**	5.09	1.85	6.48
2. Labor Market	12.25	4.66	5.88	6.62	**26.96**	3.19	17.89	10.78	5.15	6.62
3. Housing	7.75	3.1	4.65	24.03	0	**34.11**	16.28	5.43	0.78	3.88
4. Consumption	4.58	12.92	3.33	5	8.75	14.58	**23.75**	4.58	14.58	7.92
5. Money and Credit	**21.11**	6.11	2.22	1.11	15	9.44	18.33	14.44	3.33	8.89
6. Int & Exch Rates	9.52	12.7	8.73	7.14	13.49	6.75	13.49	**13.89**	11.11	3.17
7. Prices	9.09	14.02	6.44	9.85	4.92	3.41	17.42	3.41	7.95	**23.48**
8. Stock Market	3.92	9.8	0	3.92	11.76	0	**50.98**	7.84	0	11.76

This table shows the percentage of target variables in a category that a given decision rule delivers the best result in forecasting using PCA. For a given category, a decision rule that gives the best result most frequently is denoted in **bold**. All target variables in 1-, 6-, 12- and 24-month ahead DI, DIAR and DIAR-LAG forecasting are considered.

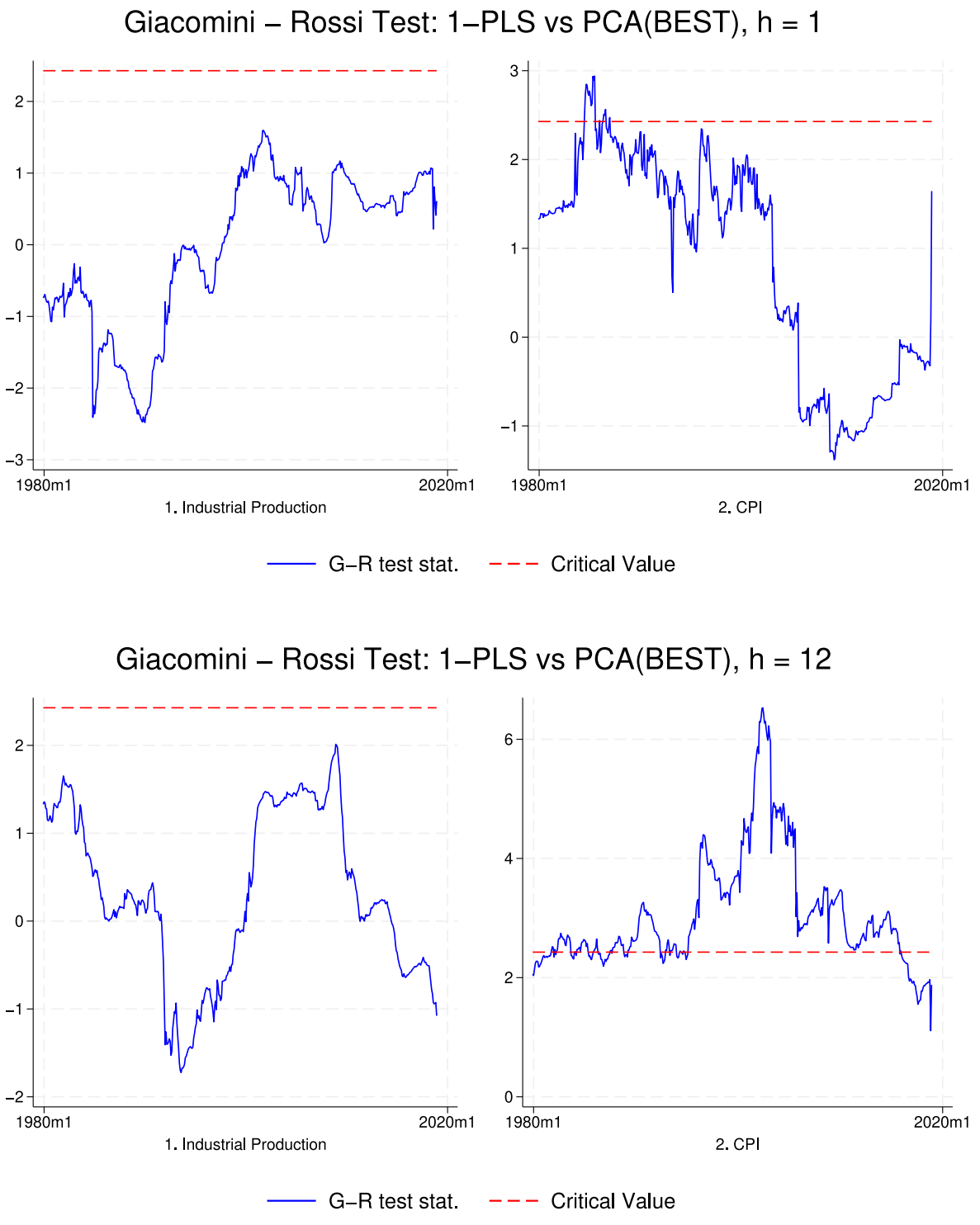
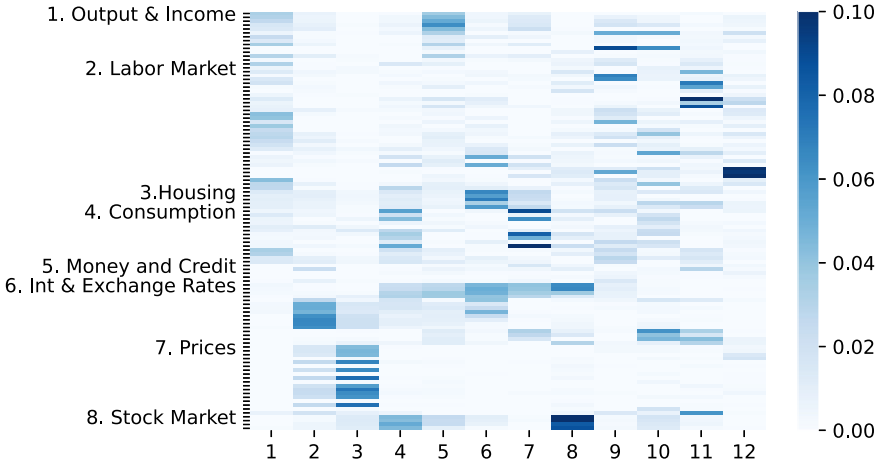
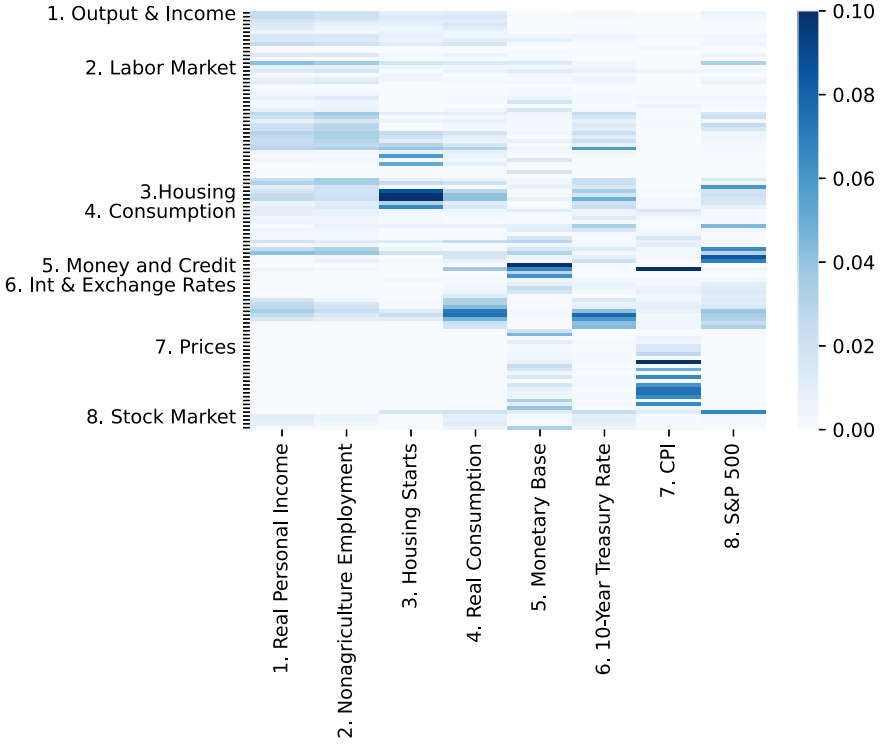


Fig. 4. Fluctuation test: 1-PLS and PCA(BEST) at $h = 1$ and $h = 12$ for 1. industrial production and 2. CPI.
Source: Giacomini and Rossi (2010).

(2022) suggest. In other words, some elements in β in the main econometric model (3) can be zero, while the corresponding factors affect the predictors x_t. Therefore, the PCA factors that explain most variations of predictors do not necessarily best predict the target variables. Incorporating weak PCA factors that explain less variation in predictors can produce better predictions for certain target variables.



(a) PCA heat map: 1-PCA to 12-PCA



(b) 1-PLS heat map: 1-PLS of each target variable

Fig. 5. Heat map of squared factor loadings: PCA and PLS.

In contrast, Table 12 shows that 1-PLS tends to be a significantly dominant decision rule for PLS. For all the eight categories, 1-PLS delivers the best forecasting performance most frequently. For most categories, 1-PLS is the best decision rule for around 60% of target variables. It is also shown that theoretically, PLS estimates the relevant factor space with priority rather than the set of factors that only govern predictors (Ahn and Bae (2022) and Kelly and Pruitt (2015)). There can be additional predictive gains as additional PLS factors are included, suggested as $\hat{k}_{ON}$, but those gains are often small in these typical macro data.

Another notable finding is that a consistently estimated number of factors may not always result in the

Table 12
Percentage of target variables that a decision rule outperforms the rest decision rules: PLS.

	1-PLS	$\hat{k}_{BIC}$	$\hat{k}_{BN-p1}$	$\hat{k}_{BN-p2}$	$\hat{k}_{BN-p3}$	$\hat{k}_{BN-BIC}$	$\hat{k}_{AH}$	$\hat{k}_{ON}$	$\hat{k}_{ABC-L}$	$\hat{k}_{ABC-S}$
1. Output & Income	**57.87**	0	0	0.46	0	20.83	7.87	9.72	1.39	1.85
2. Labor Market	**55.88**	0.49	0.98	2.21	0.49	10.29	8.09	17.4	3.43	0.74
3. Housing	**45.74**	15.5	2.33	8.53	8.53	3.88	0	6.98	6.98	1.55
4. Consumption	**67.92**	0	0.42	0	0.42	8.33	11.25	10.83	0.42	0.42
5. Money and Credit	**58.33**	1.11	0	7.78	0.56	11.11	8.33	10.56	2.22	0
6. Int & Exch Rates	**51.59**	0	0	1.19	0	4.76	10.32	31.75	0	0.4
7. Prices	**31.82**	0.38	1.14	16.67	0	24.62	4.17	13.26	6.44	1.52
8. Stock Market	**54.9**	7.84	0	7.84	0	3.92	1.96	23.53	0	0

This table shows the percentage of target variables in a category that a given decision rule delivers the best result in forecasting using PLS. For a given category, a decision rule that gives the best result most frequently is denoted in **bold**. All target variables in 1-, 6-, 12- and 24-month ahead DI, DIAR and DIAR-LAG forecasting are considered.

Table 13
Average forecasting performance of each factor estimation method by forecasting horizon and model.

		DI			DIAR			DIAR-LAG		
		Best	Worst	Mean	Best	Worst	Mean	Best	Worst	Mean
$h=1$	1-PLS	6.2	6.2	6.2	0.98	0.98	0.98	0.97	0.97	0.97
	PLS	4.74	8.42	6.28	0.96	1.79	1.29	0.96	1.26	1.11
	PCA	4.66	8.08	6.35	0.93	1	0.96	0.94	0.97	0.96
	Targeted Predictors	2.57	7.4	4.03	0.93	1.03	0.97	0.94	0.99	0.96
	Weighted PC	4.47	8.73	6.39	0.93	1.04	0.97	0.94	1.02	0.97
	Two Step	4.73	8.06	6.26	0.94	0.99	0.96	0.94	0.97	0.96
	QMLE	4.82	8.08	6.3	0.94	1	0.97	0.94	0.98	0.96
	One-sided	3.25	8.13	5.8	0.93	0.99	0.96	0.94	0.98	0.96
$h=6$	1-PLS	2.54	2.54	2.54	0.91	0.91	0.91	0.92	0.92	0.92
	PLS	2.22	3.44	2.73	0.89	1.56	1.14	0.9	1.18	1.05
	PCA	1.97	2.87	2.42	0.87	0.99	0.92	0.87	0.93	0.9
	Targeted Predictors	1.54	2.77	1.96	0.88	1.04	0.95	0.87	0.98	0.92
	Weighted PC	1.91	3.13	2.44	0.86	1.05	0.94	0.87	1.01	0.94
	Two Step	2.05	2.86	2.42	0.88	0.97	0.92	0.88	0.94	0.91
	QMLE	2.12	2.91	2.47	0.89	1.03	0.95	0.89	0.95	0.92
	One-sided	1.73	2.89	2.33	0.87	0.97	0.92	0.88	0.95	0.92
$h=12$	1-PLS	1.83	1.83	1.83	0.88	0.88	0.88	0.89	0.89	0.89
	PLS	1.66	2.78	2.09	0.86	1.68	1.16	0.86	1.21	1.04
	PCA	1.53	2.01	1.76	0.84	1.02	0.91	0.85	0.92	0.88
	Targeted Predictors	1.37	1.99	1.61	0.85	1.07	0.94	0.85	0.99	0.92
	Weighted PC	1.48	2.18	1.78	0.84	1.06	0.92	0.84	0.99	0.91
	Two Step	1.59	2	1.78	0.85	0.97	0.91	0.85	0.92	0.89
	QMLE	1.64	2.09	1.83	0.86	1.06	0.94	0.87	0.94	0.91
	One-sided	1.47	2.02	1.74	0.84	0.98	0.91	0.85	0.94	0.9
$h=24$	1-PLS	1.48	1.48	1.48	0.88	0.88	0.88	0.88	0.88	0.88
	PLS	1.32	2.51	1.75	0.85	1.8	1.18	0.84	1.2	1.02
	PCA	1.35	1.67	1.49	0.87	1.06	0.94	0.88	0.95	0.91
	Targeted Predictors	1.26	1.72	1.47	0.87	1.14	0.99	0.87	1.03	0.95
	Weighted PC	1.3	1.8	1.5	0.86	1.12	0.95	0.87	1.01	0.94
	Two Step	1.38	1.68	1.51	0.88	1.02	0.94	0.88	0.96	0.92
	QMLE	1.42	1.75	1.55	0.89	1.12	0.98	0.89	0.97	0.94
	One-sided	1.33	1.69	1.5	0.86	1.04	0.94	0.87	0.98	0.92

This table shows the average predictive performance of each factor estimation method by forecasting horizon and model. Dashed lines are used to denote 1-PLS. For a given factor estimation method, the best and worst RMSEs are calculated, as well as mean RMSE over all decision rules under each forecasting horizon and model. For $h = 1, 6, 12, 24$, Mean of best (Best), worst (Worst), and mean (Mean) over all target variables for each method are presented.

best forecasting. This is shown in Tables 11–12, and Fig. 3. For PLS, a consistently estimated number of factors often gives worse predictions than a naive 1-PLS. Table 11 shows that even for PCA, predictive gain often arises from decision rules that do not consistently estimate the true number of factors. For instance, $\hat{k}_{BIC}$ is not theoretically proven as a consistent estimator for r. $\hat{k}_{BN-BIC}$ may not be a consistent estimator, as shown by Bai and Ng (2002). $\hat{k}_{GP}$ is not a consistent estimator either, as the method proposed by Giovannelli and Proietti (2016) selects a subset of factors from all the possible N PCA components in a supervised way. This finding again supports the possible existence of irrelevant factors that do not explain the target variable. Empirically, predictive gain can often be obtained if the irrelevant factors are not used for forecasting. For instance, on average, $\hat{k}_{BN-BIC}$ is estimated as 3.3, which is smaller than the number of factors determined by consistent estimators, such as $\hat{k}_{BN-p1}$, $\hat{k}_{BN-p2}$, $\hat{k}_{BN-p3}$, $\hat{k}_{ABC-L}$, and $\hat{k}_{ABC-S}$.

Table 13 summarises the predictive ability of each factor estimation method across decision rules. First, for each target variable, the best, worst, and average forecasts

of a given factor estimation method across decision rules are collected. Each column in Table 13 is the average of the lowest, highest, and mean RMSEs of the given method over 144 (148) target variables by horizon.[3] Note that 1-PLS is displayed separately from PLS. Therefore, 1-PLS rows show the same values for all the columns. The PLS rows consider all decision rules applied to PLS, including 1-PLS.

Table 13 demonstrates several empirical features. First, for all factor models and forecasting horizons, the DI models' average RMSE is larger than one. Even the best possible decision rules for each factor estimation method produce worse forecasting results than the benchmark AR(BIC), indicated by RMSEs larger than one in the Best column. Therefore, forecasting with only estimated factors without the lagged target variable may not be more accurate than the AR forecasts in the recursive scheme.

Second, the best forecasts of all factor estimation methods are very similar given the forecasting horizon, especially once the dynamics of the target variable, such as DIAR and DIAR-LAG models, are considered. The best forecasting performance one can obtain does not vary drastically depending on the factor estimation method if a proper decision rule is chosen, as indicated by the small variation in the Best column. This is because most of the seven factor estimation methods are proven consistent estimators for the true factors up to rotation.

Third, forecasting performance varies widely among decision rules, even when using the same factor estimation method. This finding is emphasized by the large differences between the Best and Worst columns for all methods except 1-PLS. For example, in 12-months ahead DIAR forecasting, the worst PCA forecasts give, on average, 1.02 RMSE while the best possible PCA forecasts with a properly chosen decision rule significantly improve upon AR(BIC) as denoted by 0.84 in the Best column. Notably, all methods show larger differences between the Best and Worst results with longer forecasting horizons in DIAR and DIAR-LAG models.

This experiment demonstrates the sensitivity of the factor estimation methods to the decision rule in practice. This implies that the choice of the factor estimation method and decision rule is crucial in empirical forecasting. As shown by larger RMSEs than 1 in DI models and the Worst columns on DIAR and DIAR-LAG models, factor-augmented forecasting does not always guarantee better forecasting results than the traditional AR process.

Fourth, on the other hand, no factor estimation method, except PLS, exhibits dominant decision rules, as found in Fig. 3. Accordingly, the Mean columns represent the average predictive ability of the given method. Note that 1-PLS tends to show the mean value closest to the smallest mean value among all methods, especially at longer horizons and in DIAR and DIAR-LAG models, which again supports the strong forecasting performance of 1-PLS.

[3] A method could have several specifications, such as targeted predictors (λ_2), weighted PC, and one-sided estimation (static and dynamic version). Then, the best, worst, and mean are calculated considering all the specifications.

5. Conclusion

This paper investigates the empirical performance of factor-augmented forecasts, whose factors are estimated by various methods. First, 1-PLS tends to give the best forecasting performance among all possible alternatives. The strong forecasting performances of 1-PLS are found in many target variables, especially when the lagged target variable is incorporated and at longer forecasting horizons. The predictive gains of 1-PLS compared to other alternatives are significant. PLS estimates factors by considering the maximum covariance with a target variable. Therefore, the strong performance of PLS can be explained by its incorporation of past information on the target variable in its factor estimation steps.

Second, a consistently estimated number of factors does not necessarily lead to the best forecasts for many factor estimation methods, including PCA. The determined number of factors varies widely, even among decision rules that consistently estimate the true number of factors. It is difficult to find a dominant decision rule that tends to deliver the best results for a wide range of target variables for all factor estimation methods except for PLS. Also, when a proper decision rule is chosen, the best forecasts do not vary by factor estimation methods. If an appropriate decision rule is applied, the best predictions do not significantly depend on the factor estimation method. However, depending on the decision rule, the same factor estimation methods can yield somewhat different accuracy. Therefore, the factor estimation method and decision rule are important in forecasting performance.

Declaration of competing interest

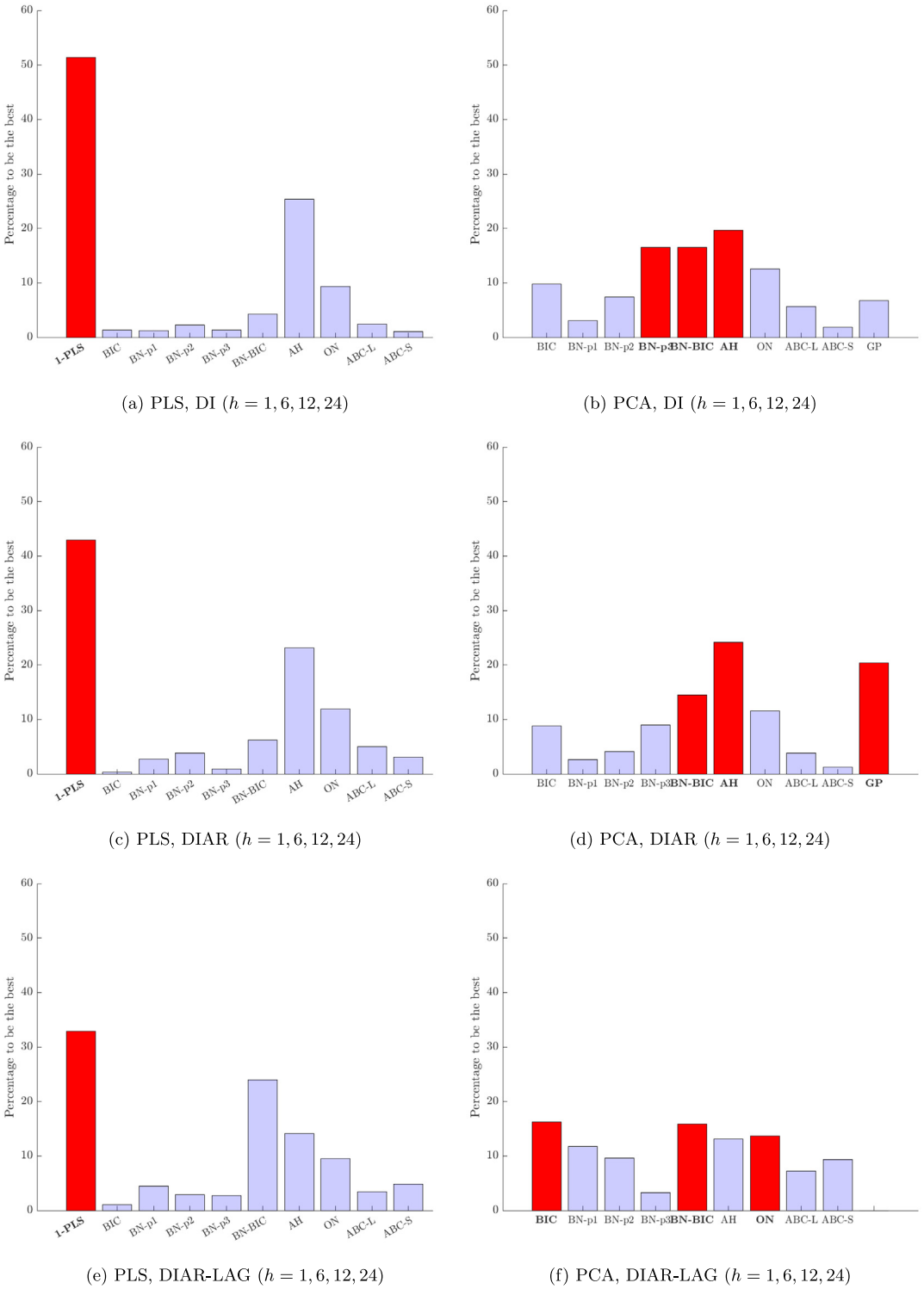
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

See Tables 14–19 and Fig. 6.

Appendix B

The variables used in this study and their categories are presented here. The tcode column denotes the transformation type. Denote the time series at t as x_t. 1 = no transformation, 2 = first difference, 3 = second difference, 4 = logarithm, 5 = first difference of logarithms 6 = second difference of logarithms, 7 = $\Delta(x_t/x_{t-1} - 1)$. The data is taken from FRED, FRED-MD (McCracken & Ng, 2016) and ISM. The source column indicates where the variable is taken from. The source is denoted as FRED if a variable belongs to both FRED and FRED-MD. If a variable in FRED-MD is adjusted from the raw data available in FRED, the source is denoted as FRED-MD. See Tables B.20–B.27.



(a) PLS, DI ($h = 1, 6, 12, 24$)

(b) PCA, DI ($h = 1, 6, 12, 24$)

(c) PLS, DIAR ($h = 1, 6, 12, 24$)

(d) PCA, DIAR ($h = 1, 6, 12, 24$)

(e) PLS, DIAR-LAG ($h = 1, 6, 12, 24$)

(f) PCA, DIAR-LAG ($h = 1, 6, 12, 24$)

Fig. 6. Percentage of Target Variables That a Decision Rule Outperform the Rest Decision Rules by Forecasting Model in **Rolling** Estimation: PLS and PCA.

Table 14
Top 5 methods in **Rolling** estimation, PLS with other decision rules excluded.

	Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
	Methods	Percentage (%)	Methods	Mean(Ratio)
$h = 1$	**1-PLS**	**16.44**	Targeted Predictors, $\lambda = 0.25$(BN-p1)	0.118
	Weighted PC, Rule B(BIC)	5.32	Targeted Predictors, $\lambda = 0.25$(BN-p2)	0.118
	QMLE(BN-p3)	5.32	Targeted Predictors, $\lambda = 0.5$(BN-p2)	0.123
	PCA(GP)	4.17	Targeted Predictors, $\lambda = 0.5$(BN-p1)	0.129
	Targeted Predictors, $\lambda = 0.25$(BN-p3)	3.94	Targeted Predictors, $\lambda = 0.5$(BIC)	0.13
			1-PLS	**0.42**
$h = 6$	**1-PLS**	**27.55**	Targeted Predictors, $\lambda = 0.25$(ABC-L)	0.167
	Weighted PC, Rule B(AH)	6.94	Targeted Predictors, $\lambda = 0.5$(ABC-L)	0.17
	Weighted PC, Rule B(BIC)	6.71	Targeted Predictors, $\lambda = 1.5$(ABC-L)	0.172
	Weighted PC, Rule B(ON)	4.17	Targeted Predictors, $\lambda = 0.5$(ON)	0.173
	PCA(BN-BIC)	3.24	Targeted Predictors, $\lambda = 0.5$(AH)	0.179
			1-PLS	**0.184**
$h = 12$	**1-PLS**	**38.19**	**1-PLS**	**0.13**
	Weighted PC, Rule B(AH)	5.32	Weighted PC, SWb(AH)	0.191
	PCA(BN-BIC)	4.17	Weighted PC, SWa(AH)	0.196
	Weighted PC, Rule B(ON)	3.47	PCA(ABC-L)	0.196
	Weighted PC, Rule B(BIC)	2.55	PCA(AH)	0.201
$h = 24$	**1-PLS**	**35.81**	**1-PLS**	**0.119**
	Weighted PC, Rule B(AH)	9.68	Weighted PC, SWa(AH)	0.182
	Weighted PC, Rule B(BIC)	6.76	OS-Static(AH)	0.187
	Weighted PC, Rule B(ON)	3.15	Weighted PC, SWb(AH)	0.187
	Weighted PC, Rule B(GP)	3.15	Weighted PC, Rule B(ON)	0.19

This table presents the top 5 methods for each forecasting horizon giving the best results most frequently and giving results closest to the best results on average in the left and right columns respectively. 1-PLS is represented in bold and dashed line represents the case that 1-PLS is not the top 5 methods. PLS with other decision rules are excluded. For each forecasting horizon, all target variables in DI, DIAR and DIAR-LAG models are considered.

Table 15
Top 10 methods : DI, DIAR and DIAR-LAG.

Recursive			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**11.15**	PCA(BIC)	0.131
PLS(BN-BIC)	5.69	OS-S(BIC)	0.131
PLS(ON)	5.57	WPC, SWb(BIC)	0.135
PLS(BN-p2)	3.33	Two-step(BIC)	0.137
OS-D(HL2007)	2.64	OS-D(BN2007)	0.138
PCA(BN-BIC)	2.59	TP, $\lambda = 1.5$(BIC)	0.14
TP, $\lambda = 1.5$(AH)	2.41	PCA(BN-p3)	0.142
WPC, SWa(ON)	2.3	WPC, SWa(BIC)	0.142
PLS(AH)	2.01	TP, $\lambda = 0.5$(BIC)	0.144
PLS(ABC-L)	1.95	OS-S(BN-p3)	0.144
		1-PLS	**0.153**
Rolling			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
PLS(AH)	14.08	**1-PLS**	**0.26**
1-PLS	**9.71**	TP, $\lambda = 0.5$(ON)	0.273
PLS(BN-BIC)	6.38	TP, $\lambda = 0.25$(ON)	0.276
WPC, Rule B(AH)	5.57	TP, $\lambda = 1.5$(ON)	0.28
PLS(ON)	5.17	PCA(ABC-L)	0.285
WPC, Rule B(BIC)	4.08	PCA(BN-BIC)	0.288
WPC, Rule B(ON)	2.64	TP, $\lambda = 0.5$(AH)	0.289
PLS(BN-p1)	2.36	TP, $\lambda = 0.25$(AH)	0.289
PLS(BN-p2)	2.3	WPC, SWb(AH)	0.29
PLS(ABC-L)	2.07	PCA(BN-p2)	0.291

This table presents the top 10 methods giving the best results most frequently and giving results closest to the best results on average according to forecasting scheme. Forecasting starts from 1970:01 and recursive scheme uses an increasing window, while rolling scheme uses a fixed window of 10 years. WPC, OS-S and OS-D stand for Weighted PC, OS-Static and OS-Dynamic. 1-PLS is represented in bold and dashed line represents the case that 1-PLS is not the top 10 methods. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

Table 16
Top 10 methods : DIAR and DIAR-LAG.

Recursive			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**10.78**	**1-PLS**	**0.081**
PLS(ON)	5.52	PCA(BN-BIC)	0.085
PLS(BN-BIC)	5.43	PCA(ABC-L)	0.085
PCA(BN-BIC)	3.19	WPC, SWb(ON)	0.085
OS-D(HL2007)	3.19	PCA(ON)	0.087
WPC, SWa(ON)	2.76	WPC, SWa(ON)	0.088
TP, $\lambda = 1.5$(AH)	2.41	PCA(AH)	0.088
QMLE(ON)	1.98	OS-S(ON)	0.089
TP, $\lambda = 0.25$(BN-BIC)	1.81	Two-step(ON)	0.09
TP, $\lambda = 1.5$(ON)	1.81	Two-step(BN-BIC)	0.091
Rolling			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
PLS(AH)	12.93	**1-PLS**	**0.216**
1-PLS	**8.97**	PLS(AH)	0.24
PLS(BN-BIC)	8.45	PLS(ON)	0.266
WPC, Rule B(AH)	6.29	TP, $\lambda = 0.5$(ON)	0.288
PLS(ON)	5.43	WPC, SWb(AH)	0.289
WPC, Rule B(ON)	3.71	TP, $\lambda = 1.5$(ON)	0.293
PLS(BN-p1)	3.28	TP, $\lambda = 0.25$(ON)	0.294
PLS(BN-p2)	2.76	PCA(AH)	0.295
PLS(ABC-S)	2.76	WPC, SWa(AH)	0.296
PLS(ABC-L)	2.67	TP, $\lambda = 0.5$(AH)	0.299

This table presents the top 10 methods giving the best results most frequently and giving results closest to the best results on average according to forecasting scheme. Forecasting starts from 1970:01 and recursive scheme uses an increasing window, while rolling scheme uses a fixed window of 10 years. WPC, OS-S and OS-D stand for Weighted PC, OS-Static and OS-Dynamic. All target variables and forecasting horizons in DIAR and DIAR-LAG forecasting models (1,160 combinations) are considered.

Table 17
Top 10 methods : DI, DIAR and DIAR-LAG, PLS with other decision rules excluded.

Recursive			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**16.55**	PCA(BIC)	0.114
PCA(BN-BIC)	3.05	OS-S(BIC)	0.115
OS-D(HL2007)	2.7	WPC, SWb(BIC)	0.118
TP, $\lambda = 1.5$(AH)	2.64	OS-D(BN2007)	0.12
WPC, SWa(ON)	2.59	Two-step(BIC)	0.121
WPC, Rule B(GP)	2.41	TP, $\lambda = 1.5$(BIC)	0.124
WPC, Rule B(BIC)	2.18	PCA(BN-p3)	0.125
TP, $\lambda = 1.5$(ON)	2.13	WPC, SWa(BIC)	0.125
TP, $\lambda = 0.5$(AH)	1.84	PCA(BN-p1)	0.127
TP, $\lambda = 1.5$(BN-p1)	1.78	PCA(BN-p2)	0.127
		1-PLS	**0.137**
Rolling			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**29.54**	**1-PLS**	**0.213**
WPC, Rule B(AH)	6.49	TP, $\lambda = 0.5$(ON)	0.219
WPC, Rule B(BIC)	5.34	TP, $\lambda = 0.25$(ON)	0.221
WPC, Rule B(ON)	2.93	TP, $\lambda = 1.5$(ON)	0.225
PCA(BN-BIC)	2.53	PCA(ABC-L)	0.23
WPC, Rule B(GP)	2.07	PCA(BN-BIC)	0.232
WPC, Rule B(BN-BIC)	1.72	TP, $\lambda = 0.5$(AH)	0.234
TP, $\lambda = 0.25$(BN-p3)	1.67	TP, $\lambda = 0.25$(AH)	0.234
TP, $\lambda = 1.5$(AH)	1.61	PCA(BN-p2)	0.235
QMLE(BN-p3)	1.55	WPC, SWb(AH)	0.235

This table presents the top 10 methods giving the best results most frequently and giving results closest to the best results on average according to forecasting scheme. Forecasting starts from 1970:01 and recursive scheme uses an increasing window, while rolling scheme uses a fixed window of 10 years. WPC, OS-S and OS-D stand for Weighted PC, OS-Static and OS-Dynamic. PLS with other decision rules are excluded. 1-PLS is represented in bold and dashed line represents the case that 1-PLS is not the top 10 methods. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

Table 18
Top 10 methods : DIAR and DIAR-LAG, PLS with other decision rules excluded.

Recursive			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**16.64**	**1-PLS**	**0.072**
PCA(BN-BIC)	3.71	PCA(BN-BIC)	0.074
OS-D(HL2007)	3.28	PCA(ABC-L)	0.074
WPC, SWa(ON)	3.19	WPC, SWb(ON)	0.075
TP, $\lambda = 1.5$(AH)	2.59	PCA(ON)	0.076
TP, $\lambda = 1.5$(ON)	2.16	PCA(AH)	0.078
WPC, SWa(AH)	2.07	WPC, SWa(ON)	0.078
QMLE(ON)	1.98	OS-S(ON)	0.079
TP, $\lambda = 0.25$(BN-BIC)	1.9	Two-step(ON)	0.08
PCA(ON)	1.81	Two-step(BN-BIC)	0.08
Rolling			
Methods Giving the Best Results Most Frequently		Methods Giving Results Closest to the Best	
Methods	Percentage (%)	Methods	Mean(Ratio)
1-PLS	**32.93**	**1-PLS**	**0.156**
WPC, SWb(ON)	7.67	WPC, SWa(ON)	0.219
WPC, SWb(ABC-L)	4.14	TP, $\lambda = 0.25$(ABC-L)	0.22
PCA(BN-BIC)	2.41	TP, $\lambda = 0.5$(ABC-L)	0.225
WPC, SWb(AH)	2.41	PLS(AH)	0.225
TP, $\lambda = 0.5$(ON)	1.72	PCA(ABC-L)	0.226
WPC, SWb(BN-p1)	1.72	TP, $\lambda = 1.5$(ON)	0.226
PLS(BN-BIC)	1.64	WPC, Rule B(ON)	0.229
Two-step(BN-p3)	1.64	TP, $\lambda = 0.25$(ON)	0.23
WPC, SWb(ABC-S)	1.47	PLS(ABC-L)	0.232

This table presents the top 10 methods giving the best results most frequently and giving results closest to the best results on average according to forecasting scheme. Forecasting starts from 1970:01 and recursive scheme uses an increasing window, while rolling scheme uses a fixed window of 10 years. WPC, OS-S and OS-D stand for Weighted PC, OS-Static and OS-Dynamic. PLS with other decision rules are excluded. All target variables and forecasting horizons in DIAR and DIAR-LAG forecasting models (1,160 combinations) are considered.

Table 19
Top 10 methods giving significant DM statistics most frequently by significance level (α).

	Recursive		Rolling	
	Method	Percentage (%)	Method	Percentage (%)
$\alpha = 0.1$	WPC, SWb(ON)	38.74	**1-PLS**	**47.3**
	PCA(ON)	38.45	PLS(AH)	42.47
	WPC, SWa(ON)	38.28	PLS(ON)	40.11
	OS-S(ON)	37.01	WPC, SWb(AH)	37.53
	Two-step(ON)	36.55	WPC, SWa(AH)	37.47
	PCA(AH)	36.03	OS-S(AH)	36.9
	WPC, SWb(AH)	34.66	PCA(AH)	36.78
	QMLE(ON)	34.6	Two-step(AH)	36.32
	1-PLS	**34.31**	QMLE(AH)	35.98
	WPC, SWa(AH)	33.97	WPC, Rule B(AH)	35.4
$\alpha = 0.05$	PCA(ON)	27.93	**1-PLS**	**39.2**
	WPC, SWb(ON)	27.24	PLS(AH)	36.15
	WPC, SWa(ON)	25.92	PLS(ON)	34.2
	OS-S(ON)	25.8	WPC, SWb(AH)	31.84
	PCA(AH)	25.4	WPC, SWa(AH)	31.38
	Two-step(ON)	25.4	OS-S(AH)	30.92
	1-PLS	**24.66**	PCA(AH)	30.69
	PCA(BN-BIC)	24.66	WPC, SWb(ON)	30.23
	QMLE(ON)	23.51	Two-step(AH)	29.77
	OS-S(AH)	23.51	WPC, SWa(ON)	29.37

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Table 19 (*continued*).

	Recursive		Rolling	
	Method	Percentage (%)	Method	Percentage (%)
$\alpha = 0.01$	OS-S(ON)	11.9	**1-PLS**	**27.93**
	WPC, SWb(ON)	11.84	PLS(ON)	26.44
	PCA(BN-BIC)	11.72	PLS(AH)	25.86
	PCA(ON)	11.44	WPC, SWb(ON)	22.36
	1-PLS	**11.09**	WPC, SWa(AH)	21.95
	PCA(AH)	10.98	WPC, SWb(AH)	21.84
	WPC, SWa(ON)	10.98	WPC, SWa(ON)	21.72
	Two-step(BN-BIC)	10.8	PCA(ON)	21.15
	Two-step(ON)	10.8	PCA(AH)	20.98
	Two-step(AH)	10.57	Two-step(AH)	20.69

This table presents the top 10 methods giving significant DM statistics most frequently, according to significance level (α) and forecasting scheme. Forecasting starts from 1970:01 and recursive scheme uses an increasing window, while rolling scheme uses a fixed window of 10 years. WPC, OS-S and OS-D stand for Weighted PC, OS-Static and OS-Dynamic. Note that the case with smaller α is also counted in larger α. For instance, if a DM statistic is significant under $\alpha = 0.05$, it is also counted in $\alpha = 0.1$. All target variables, forecasting horizons and forecasting models (1,740 combinations) are considered.

Table B.20
Category 1: Output and Income.

ID	Variable	tcode	Description	Period	Target	Source
1	RPI	5	Real Personal Income	1959:01–2019:12	1, 6, 12, 24	FRED
2	W875RX1	5	Real personal income excl. transfer receipts	1959:01–2019:12	1, 6, 12, 24	FRED
6	INDPRO	5	IP: Total Index	1959:01–2019:12	1, 6, 12, 24	FRED
7	IPFPNSS	5	IP: Final Products and Nonindustrial Supplies	1959:01–2019:12	1, 6, 12, 24	FRED
8	IPFINAL	5	IP: Final Products	1959:01–2019:12	1, 6, 12, 24	FRED
9	IPCONGD	5	IP: Consumer Goods	1959:01–2019:12	1, 6, 12, 24	FRED
10	IPDCONGD	5	IP: Durable Consumer Goods	1959:01–2019:12	1, 6, 12, 24	FRED
11	IPNCONGD	5	IP: Nondurable Consumer Goods	1959:01–2019:12	1, 6, 12, 24	FRED
12	IPBUSEQ	5	IP: Equipment: Business Equipment	1959:01–2019:12	1, 6, 12, 24	FRED
13	IPMAT	5	IP: Materials	1959:01–2019:12	1, 6, 12, 24	FRED
14	IPDMAT	5	IP: Durable Goods Materials	1959:01–2019:12	1, 6, 12, 24	FRED
15	IPNMAT	5	IP: Nondurable Goods Materials	1959:01–2019:12	1, 6, 12, 24	FRED
16	IPMANSICS	5	IP: Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
17	IPB51222S	5	IP: Utilities	1959:01–2019:12	1, 6, 12, 24	FRED
18	IPFUELS	5	IP: Fuels	1959:01–2019:12	1, 6, 12, 24	FRED
19	CUMFNS	2	Capacity Util: Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
129	IPNMAN	5	IP: Nondurable Manufacturing	1972:01–2019:12		FRED
130	IPDMAN	5	IP: Durable Manufacturing	1972:01–2019:12		FRED
131	IPMINE	5	IP: Mining	1959:01–2019:12	1, 6, 12, 24	FRED
132	TCU	1	Capacity Util: Total Index	1967:01–2019:12		FRED
133	CAPUTLGMFDS	1	Capacity Util: Durable Manufacturing	1967:01–2019:12		FRED
134	CAPUTLGMFNS	1	Capacity Util: Nondurable Manufacturing	1967:01–2019:12		FRED
135	CAPUTLG21S	1	Capacity Util: Mining	1967:01–2019:12		FRED
136	CAPUTLG2211A2S	1	Capacity Util: Utilities	1967:01–2019:12		FRED
173	ISM \MAN_PROD	1	Manufacturing Production Index	1959:01–2019:12	1, 6, 12, 24	ISM

Table B.21
Category 2: Labor market.

ID	Variable	tcode	Description	Period	Target	Source
20	HWI	2	Help-Wanted Index for US	1959:01–2019:12	1, 6, 12, 24	FRED-MD
21	HWIURATIO	2	Ratio of Help Wanted/No. Unemployed	1959:01–2019:12	1, 6, 12, 24	FRED-MD
22	CLF16OV	5	Civilian Labor Force Level	1959:01–2019:12	1, 6, 12, 24	FRED
23	CE16OV	5	Employment Level	1959:01–2019:12	1, 6, 12, 24	FRED
24	UNRATE	2	Unemployment Rate	1959:01–2019:12	1, 6, 12, 24	FRED
25	UEMPMEAN	2	Avg Weeks Unemployed	1959:01–2019:12	1, 6, 12, 24	FRED
26	UEMPLT5	5	Number Unemployed for Less Than 5 Weeks	1959:01–2019:12	1, 6, 12, 24	FRED
27	UEMP5TO14	5	Number Unemployed for 5-14 Weeks	1959:01–2019:12	1, 6, 12, 24	FRED
28	UEMP15OV	5	Number Unemployed for 15 Weeks & over	1959:01–2019:12	1, 6, 12, 24	FRED
29	UEMP15T26	5	Number Unemployed for 15-26 Weeks	1959:01–2019:12	1, 6, 12, 24	FRED
30	UEMP27OV	5	Number Unemployed for 27 Weeks & over	1959:01–2019:12	1, 6, 12, 24	FRED
31	CLAIMSx	5	Initial Claims	1959:01–2019:12	1, 6, 12, 24	FRED-MD
32	PAYEMS	5	All Employees, Total Nonfarm	1959:01–2019:12	1, 6, 12, 24	FRED
33	USGOOD	5	All Employees, Goods-Producing	1959:01–2019:12	1, 6, 12, 24	FRED
34	CES1021000001	5	All Employees, Mining, Quarrying, and Oil and Gas Extraction	1959:01–2019:12	1, 6, 12, 24	FRED
35	USCONS	5	All Employees, Construction	1959:01–2019:12	1, 6, 12, 24	FRED

(*continued on next page*)

Table B.21 (*continued*).

ID	Variable	tcode	Description	Period	Target	Source
36	MANEMP	5	All Employees, Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
37	DMANEMP	5	All Employees, Durable Goods	1959:01–2019:12	1, 6, 12, 24	FRED
38	NDMANEMP	5	All Employees, Nondurable Goods	1959:01–2019:12	1, 6, 12, 24	FRED
39	SRVPRD	5	All Employees, Service-Providing	1959:01–2019:12	1, 6, 12, 24	FRED
40	USTPU	5	All Employees, Trade, Trans. & Utilities	1959:01–2019:12	1, 6, 12, 24	FRED
41	USWTRADE	5	All Employees, Wholesale Trade	1959:01–2019:12	1, 6, 12, 24	FRED
42	USTRADE	5	All Employees, Retail Trade	1959:01–2019:12	1, 6, 12, 24	FRED
43	USFIRE	5	All Employees, Financial Activities	1959:01–2019:12	1, 6, 12, 24	FRED
44	USGOVT	5	All Employees, Government	1959:01–2019:12	1, 6, 12, 24	FRED
45	CES0600000007	1	Avg Weekly Hours, Goods-Producing	1959:01–2019:12	1, 6, 12, 24	FRED
46	AWOTMAN	2	Avg Weekly Overtime Hours, Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
47	AWHMAN	1	Avg Weekly Hours, Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
120	CES0600000008	6	Avg Hourly Earnings, Goods-Producing	1959:01–2019:12	1, 6, 12, 24	FRED
121	CES2000000008	6	Avg Hourly Earnings, Construction	1959:01–2019:12	1, 6, 12, 24	FRED
122	CES3000000008	6	Avg Hourly Earnings, Manufacturing	1959:01–2019:12	1, 6, 12, 24	FRED
137	USPRIV	5	All Employees, Total Private	1959:01–2019:12	1, 6, 12, 24	FRED
138	CES5552000001	5	All Employees, Finance and Insurance	1990:01–2019:12		FRED
139	CES5553100001	5	All Employees, Real Estate	1990:01–2019:12		FRED
140	SRVPRD	5	All Employees, Service-Providing	1959:01–2019:12	1, 6, 12, 24	FRED
141	AWHNONAG	1	Avg Weekly Hours, Total Private	1964:01–2019:12		FRED
166	AHETPI	6	Avg Hourly Earnings, Total Private	1964:01–2019:12		FRED
167	CES4000000008	6	Avg Hourly Earnings, Trade, Trans. & Utilities	1964:01–2019:12		FRED
168	CES4200000008	6	Avg Hourly Earnings, Retail Trade	1972:01–2019:12		FRED
169	CES4142000008	6	Avg Hourly Earnings, Wholesale Trade	1972:01–2019:12		FRED
170	CES5500000008	6	Avg Hourly Earnings, Financial Activities	1964:01–2019:12		FRED
171	CES0800000008	6	Avg Hourly Earnings, Private Service-Providing	1964:01–2019:12		FRED
174	ISM \MAN_EMPL	1	Manufacturing Employment Index	1959:01–2019:12	1, 6, 12, 24	ISM

Table B.22
Category 3: Housing.

ID	Variable	tcode	Description	Period	Target	Source
48	HOUST	4	Housing Starts: Total	1959:01–2019:12	1, 6, 12, 24	FRED
49	HOUSTNE	4	Housing Starts: Northeast	1959:01–2019:12	1, 6, 12, 24	FRED
50	HOUSTMW	4	Housing Starts: Midwest	1959:01–2019:12	1, 6, 12, 24	FRED
51	HOUSTS	4	Housing Starts: South	1959:01–2019:12	1, 6, 12, 24	FRED
52	HOUSTW	4	Housing Starts: West	1959:01–2019:12	1, 6, 12, 24	FRED
53	PERMIT	4	New Private Housing Permits: Total	1960:01–2019:12	1, 6, 12, 24	FRED
54	PERMITNE	4	New Private Housing Permits: Northeast	1960:01–2019:12	1, 6, 12, 24	FRED
55	PERMITMW	4	New Private Housing Permits: Midwest	1960:01–2019:12	1, 6, 12, 24	FRED
56	PERMITS	4	New Private Housing Permits: South	1960:01–2019:12	1, 6, 12, 24	FRED
57	PERMITW	4	New Private Housing Permits: West	1960:01–2019:12	1, 6, 12, 24	FRED
150	HSN1F	4	New One Family Houses Sold: US	1963:01–2019:12	24	FRED
151	HSN1FNE	4	New One Family Houses Sold, Northeast	1973:01–2019:12		FRED
152	HSN1FMW	4	New One Family Houses Sold, Midwest	1973:01–2019:12		FRED
153	HSN1FS	4	New One Family Houses Sold, South	1973:01–2019:12		FRED
154	HSN1FW	4	New One Family Houses Sold, West	1973:01–2019:12		FRED
155	MSACSR	4	Monthly Supply of New Houses, US	1963:01–2019:12	24	FRED
156	HNFSEPUSSA	4	New One Family Homes for Sale, US	1963:01–2019:12	24	FRED
157	UNDCONTSA	4	New Privately-Owned Housing Under Construction: Total	1970:01–2019:12		FRED

Table B.23
Category 4: Consumption, orders, and inventories.

ID	Variable	tcode	Description	Period	Target	Source
3	DPCERA3M086SBEA	5	Real personal consumption expenditures	1959:01–2019:12	1, 6, 12, 24	FRED
4	CMRMTSPLx	5	Real Manu. and Trade Industries Sales	1959:01–2019:12	1, 6, 12, 24	FRED-MD
5	RETAILx	5	Retail and Food Services Sales	1959:01–2019:12	1, 6, 12, 24	FRED-MD
58	ACOGNO	5	Manufacturers' New Orders: Consumer Goods	1992:02-2019:12		FRED
59	AMDMNOx	5	New Orders for Durable Goods	1959:01–2019:12	1, 6, 12, 24	FRED-MD
60	ANDENOx	5	New Orders for Nondefense Capital Goods	1968:02-2019:12		FRED-MD
61	AMDMUOx	5	Unfilled Orders for Durable Goods	1959:01–2019:12	1, 6, 12, 24	FRED-MD
62	BUSINVx	5	Total Business Inventories	1959:01–2019:12	1, 6, 12, 24	FRED-MD
63	ISRATIOx	2	Total Business: Inventories to Sales Ratio	1959:01–2019:12	1, 6, 12, 24	FRED-MD
123	UMCSENTx	2	Consumer Sentiment Index	1959:05-2019:12		FRED-MD
142	USASLMNTO02MLSAM	5	Sales: Manufacturing: Total Manufacturing: Value for US	1960:01–2019:12	1, 6, 12, 24	FRED
143	USASLRTTO02MLSAM	5	Sales: Retail Trade: Total Retail Trade: Value for US	1960:01–2019:12	1, 6, 12, 24	FRED
144	USASLWHTO02MLSAM	5	Sales: Wholesale Trade: Total Wholesale Trade: Value for US	1960:01–2019:12	1, 6, 12, 24	FRED
145	USASARTMISMEI	1	Total Retail Trade in US	1960:01–2019:12	1, 6, 12, 24	FRED
146	DDURRA3M086SBEA	5	Real pers. consump. expnd.: Durable goods	1959:01–2019:12	1, 6, 12, 24	FRED
147	DNDGRA3M086SBEA	5	Real pers. consump. expnd.: Nondurable goods	1959:01–2019:12	1, 6, 12, 24	FRED
148	DSERRA3M086SBEA	5	Real pers. consump. expnd.: Services	1959:01–2019:12	1, 6, 12, 24	FRED
149	DGDSRA3M086SBEA	5	Real pers. consump. expnd.: Goods	1959:01–2019:12	1, 6, 12, 24	FRED

(*continued on next page*)

Table B.23 (*continued*).

ID	Variable	tcode	Description	Period	Target	Source
158	INVCMRMTSPL	5	Real Manufacturing and Trade Inventories	1967:01–2019:12		FRED
159	SOANDI	1	CFNAI: Sales, Orders and Inventories	1967:03-2019:12		FRED
160	USAODMNTO02MLSAM	5	Orders: Manufacturing: Total Orders: Value for US	1960:01–2019:12	1, 6, 12, 24	FRED
172	ISM \MAN_PMI	1	PMI Composite Index	1959:01–2019:12	1, 6, 12, 24	ISM
175	ISM \MAN_NEWORDERS	1	Manufacturing New Orders Index	1959:01–2019:12	1, 6, 12, 24	ISM
176	ISM \MAN_DELIV	1	Manufacturing Supplier Deliveries Index	1959:01–2019:12	1, 6, 12, 24	ISM
177	ISM \MAN_INVENT	1	Manufacturing Inventories Index	1959:01–2019:12	1, 6, 12, 24	ISM

Table B.24
Category 5: Money and credit.

ID	Variable	tcode	Description	Period	Target	Source
64	M1SL	6	M1	1959:01–2019:12	1, 6, 12, 24	FRED
65	M2SL	6	M2	1959:01–2019:12	1, 6, 12, 24	FRED
66	M2REAL	5	Real M2 Money Stock	1959:01–2019:12	1, 6, 12, 24	FRED
67	BOGMBASE	6	Monetary Base; Total	1959:01–2019:12	1, 6, 12, 24	FRED
68	TOTRESNS	6	Reserves of Depository Institutions: Total	1959:01–2019:12	1, 6, 12, 24	FRED
69	NONBORRES	7	Reserves of Depository Institutions, Nonborrowed	1959:01–2019:12	1, 6, 12, 24	FRED
70	BUSLOANS	6	Commercial and Industrial Loans	1959:01–2019:12	1, 6, 12, 24	FRED
71	REALLN	6	Real Estate Loans, All Commercial Banks	1959:01–2019:12	1, 6, 12, 24	FRED
72	NONREVSL	6	Total Nonrevolving Credit	1959:01–2019:12	1, 6, 12, 24	FRED
73	CONSPI	2	Nonrevolving consumer credit to Personal Income	1959:01–2019:12	1, 6, 12, 24	FRED-MD
124	MZMSL	6	MZM Money Stock	1959:01–2019:12	1, 6, 12, 24	FRED
125	DTCOLNVHFNM	6	Consumer Motor Vehicle Loans Outstanding	1959:01–2019:12	1, 6, 12, 24	FRED
126	DTCTHFNM	6	Total Consumer Loans and Leases Outstanding	1959:01–2019:12	1, 6, 12, 24	FRED
127	INVEST	6	Securities in Bank Credit, All Commercial Banks	1959:01–2019:12	1, 6, 12, 24	FRED
162	USGSEC	5	Treasury and Agency Securities, All Commercial Banks	1959:01–2019:12	1, 6, 12, 24	FRED

Table B.25
Category 6: Interest and exchange rates.

ID	Variable	tcode	Description	Period	Target	Source
78	FEDFUNDS	2	Federal Funds Effective Rate	1959:01–2019:12	1, 6, 12, 24	FRED
79	CP3Mx	2	3-Month AA Financial Commercial Paper Rate	1959:01–2019:12	1, 6, 12, 24	FRED-MD
80	TB3MS	2	3-Month Treasury Bill , Discount Basis	1959:01–2019:12	1, 6, 12, 24	FRED
81	TB6MS	2	6-Month Treasury Bill , Discount Basis	1959:01–2019:12	1, 6, 12, 24	FRED
82	GS1	2	1-Year Treasury Rate, Quoted on an Investment Basis	1959:01–2019:12	1, 6, 12, 24	FRED
83	GS5	2	5-Year Treasury Rate, Quoted on an Investment Basis	1959:01–2019:12	1, 6, 12, 24	FRED
84	GS10	2	10-Year Treasury Rate, Quoted on an Investment Basis	1959:01–2019:12	1, 6, 12, 24	FRED
85	AAA	2	Moody's Seasoned Aaa Corporate Bond Yield	1959:01–2019:12	1, 6, 12, 24	FRED
86	BAA	2	Moody's Seasoned Baa Corporate Bond Yield	1959:01–2019:12	1, 6, 12, 24	FRED
87	COMPAPFFx	1	3-Month Commercial Paper Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED-MD
88	TB3SMFFM	1	3-Month Treasury Bill Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
89	TB6SMFFM	1	6-Month Treasury Bill Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
90	T1YFFM	1	1-Year Treasury Constant Maturity Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
91	T5YFFM	1	5-Year Treasury Constant Maturity Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
92	T10YFFM	1	10-Year Treasury Constant Maturity Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
93	AAAFFM	1	Moody's Seasoned Aaa Corporate Bond Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
94	BAAFFM	1	Moody's Seasoned Baa Corporate Bond Minus FEDFUNDS	1959:01–2019:12	1, 6, 12, 24	FRED
95	TWEXAFEGSMTHx	5	Trade Weighted U.S. Dollar Index	1973:01–2019:12		FRED-MD
96	EXSZUSx	5	Switzerland/U.S. Foreign Exchange Rate	1959:01–2019:12	1, 6, 12, 24	FRED-MD
97	EXJPUSx	5	Japan/U.S. Foreign Exchange Rate	1959:01–2019:12	1, 6, 12, 24	FRED-MD
98	EXUSUKx	5	U.S./U.K. Foreign Exchange Rate	1959:01–2019:12	1, 6, 12, 24	FRED-MD
99	EXCAUSx	5	Canada/U.S. Foreign Exchange Rate	1959:01–2019:12	1, 6, 12, 24	FRED-MD
161	RNUSBIS	5	Real Narrow Effective Exchange Rate for US	1964:01–2019:12		FRED

Table B.26
Category 7: Prices.

ID	Variable	tcode	Description	Period	Target	Source
100	WPSFD49207	6	PPI: Finished Goods	1959:01–2019:12	1, 6, 12, 24	FRED
101	WPSFD49502	6	PPI: Finished Consumer Goods	1959:01–2019:12	1, 6, 12, 24	FRED
102	WPSID61	6	PPI: Processed Intermediate Goods	1959:01–2019:12	1, 6, 12, 24	FRED
103	WPSID62	6	PPI: Unprocessed Intermediate Goods	1959:01–2019:12	1, 6, 12, 24	FRED
104	OILPRICEx	6	Crude Oil, spliced WTI and Cushing	1959:01–2019:12	1, 6, 12, 24	FRED-MD
105	PPICMM	6	PPI: Metals and Metal Products	1959:01–2019:12	1, 6, 12, 24	FRED
106	CPIAUCSL	6	CPI: All Items	1959:01–2019:12	1, 6, 12, 24	FRED
107	CPIAPPSL	6	CPI: Apparel	1959:01–2019:12	1, 6, 12, 24	FRED
108	CPITRNSL	6	CPI: Transportation	1959:01–2019:12	1, 6, 12, 24	FRED
109	CPIMEDSL	6	CPI: Medical Care	1959:01–2019:12	1, 6, 12, 24	FRED
110	CUSR0000SAC	6	CPI: Commodities	1959:01–2019:12	1, 6, 12, 24	FRED

(*continued on next page*)

Table B.26 (*continued*).

ID	Variable	tcode	Description	Period	Target	Source
111	CUSR0000SAD	6	CPI: Durables	1959:01–2019:12	1, 6, 12, 24	FRED
112	CUSR0000SAS	6	CPI: Services	1959:01–2019:12	1, 6, 12, 24	FRED
113	CPIULFSL	6	CPI: All Items Less Food	1959:01–2019:12	1, 6, 12, 24	FRED
114	CUSR0000SA0L2	6	CPI: All Items Less Shelter	1959:01–2019:12	1, 6, 12, 24	FRED
115	CUSR0000SA0L5	6	CPI: All Items Less Medical Care	1959:01–2019:12	1, 6, 12, 24	FRED
116	PCEPI	6	PCE: Chain Index	1959:01–2019:12	1, 6, 12, 24	FRED
117	DDURRG3M086SBEA	6	PCE: Durable goods	1959:01–2019:12	1, 6, 12, 24	FRED
118	DNDGRG3M086SBEA	6	PCE: Nondurable goods	1959:01–2019:12	1, 6, 12, 24	FRED
119	DSERRG3M086SBEA	6	PCE: Services	1959:01–2019:12	1, 6, 12, 24	FRED
163	WPSFD49209	6	PPI: Finished Goods, Excluding Foods	1967:01–2019:12		FRED
164	CPIUFDSL	6	CPI: Food	1959:01–2019:12	1, 6, 12, 24	FRED
165	CPIHOSSL	6	CPI: Housing	1967:01–2019:12		FRED
178	ISM \MAN_PRICES	1	Manufacturing Prices Index	1959:01–2019:12	1, 6, 12, 24	ISM

Table B.27
Category 8: Stock market.

ID	Variable	tcode	Description	Period	Target	Source
74	SP_500	5	S&P's Common Stock Price Index: Composite	1959:01–2019:12	1, 6, 12, 24	FRED-MD
75	SP_indust	5	S&P's Common Stock Price Index: Industrials	1959:01–2019:12	1, 6, 12, 24	FRED-MD
76	SP_div_yield	2	S&P's Composite Common Stock: Dividend Yield	1959:01–2019:12	1, 6, 12, 24	FRED-MD
77	SP_PE_ratio	5	S&P's Composite Common Stock: Price-Earnings Ratio	1959:01–2019:12	1, 6, 12, 24	FRED-MD
128	VXOCLSx	1	VXO	1962:07-2019:12	24	FRED-MD

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